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Optimizing biochar use in rice production for improving soil health and environmental sustainability: A global meta-analysis guided insights

Alok Sinha [a], C.M. Parihar [a,*], P. Panneerselvam [b,*], Ayan Sarkar [a], K. Srikanth Reddy [a], Kiranmoy Patra [a], Sneha Bharadwaj [a], Madhumita Patra [c], D.K. Sharma [a], S.L. Jat [d], S.S. Rathore [a], Sunil Kumar [b], Sonam R. Sherpa [e], H.S. Nayak [a,f], Virender Kumar [g]

[a] ICAR-Indian Agricultural Research Institute (IARI), New Delhi, India
[b] International Rice Research Institute (IRRI), South Asia Regional Centre, Varanasi, India
[c] CIPMC, Bhubaneswar, India
[d] ICAR-Indian Institute of Maize Research, Ludhaina, Punjab, India
[e] International Maize and Wheat Improvement Center (CIMMYT), Nepal
[f] Cornell University, Ithaca, NY, USA
[g] International Rice Research Institute (IRRI), Los Banos, Philippines

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ABSTRACT

Rice cultivation plays a pivotal role in global food security but remains a major source of greenhouse gas (GHG) emissions, particularly methane (CH_4) and nitrous oxide (N_2O). To tackle this, several researchers have shown the potential of biochar, a carbon-rich material derived from biomass pyrolysis, in lowering emissions, optimizing yield emission trade-offs, and enhancing soil health. However, most of these studies are highly region and context-specific. Therefore, a robust and comprehensive, global-scale investigation is important to quantify the actual impact of biochar on rice yield, emission trade-offs, and soil health. This meta-analysis systematically evaluated the effects of biochar application on GHG emissions, global warming potential (GWP), greenhouse gas intensity (GHGI), total soil carbon stock response (TSCSR), and rice yield. From an initial pool of 27,000 publications, 54 studies were selected based on strict inclusion criteria, contributing a total of 1143 paired observations: 197 for CH_4, 164 for N_2O, 176 for GWP, 176 for GHGI, 336 for grain yield, and 94 for TSCSR. On average, biochar application was associated with reductions in CH_4 (18.11%) and N_2O (15.54%) emissions, along with decreases in GWP (25.51%) and GHGI (33.16%). Additionally, TSCSR increased by 69.96%, and rice yield increased by 11.13%. However, the increase in TSCSR may, to some extent, reflect the direct contribution of biochar-derived carbon, along with changes in native soil organic carbon. The magnitude of these responses varied across climatic conditions, soil properties, management practices, and biochar properties. Subtropical regions showed consistent reductions across all studied variables, whereas tropical regions exhibited the strongest reductions in integrated indices (GWP and GHGI). Across soil textures, emission reductions (CH_4, GWP, and GHGI) were more pronounced in coarse-textured soils, while yield and TSCSR responses were greater in fine-textured soils. Stronger responses were also observed in acidic soils with moderate organic carbon and low total nitrogen. Additionally, biochar produced from high C:N ratio feedstocks, with an acidic pH, and generated at lower pyrolysis temperatures (<650°C), showed a comparatively stronger response. Puddled transplanted rice systems with continuous flooding and high nitrogen inputs typically enhance CH_4 emissions. However, biochar application at moderate rates (10–30 t ha^{-1}) was associated with reduced GHG emissions and improved productivity. These findings emphasize biochars potential to support sustainable rice production by reducing emissions and improving yield. Future efforts should focus on large-scale, long-term trials and incentivizing adoption through carbon credit frameworks to advance climate-resilient agriculture.

Abbreviation: CH_4, Methane; N_2O, Nitrous oxide; GHG, Greenhouse gas; GWP, Global warming potential; GHGI, Greenhouse gas intensity; TSCSR, Total soil carbon stock response; SOC, Soil organic carbon; TN, Total nitrogen.

* Corresponding authors.
E-mail addresses: cmparihar@iari.res.in (C.M. Parihar), p.panneerselvam@irri.org (P. Panneerselvam).

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1. Introduction

The global population has reached 8.2 billion and is projected to peak at approximately 10.3 billion by the mid-2080s (WPP, 2024), placing considerable pressure on agriculture to meet rising food demand. Rice (*Oryza sativa*) plays a vital role in this context, serving as the staple food for more than 3.5 billion people and providing nearly 20% of the global dietary energy supply (Deng et al., 2024; Wen et al., 2025). However, conventional rice production faces increasing challenges due to climate change, resource limitations, and environmental degradation (Liu et al., 2024). Climate change remains one of the most pressing global challenges (Reddy et al., 2025), with current estimates indicating that global warming has reached 1.1°C above pre-industrial levels (IPCC et al., 2023). The Paris Agreement seeks to limit this rise to below 1.5°C or 2°C (Roebroek et al., 2023), necessitating a significant reduction in greenhouse gas (GHG) emissions. Agriculture accounts for nearly 25% of global GHG emissions (Crippa et al., 2021), primarily from livestock, rice cultivation, and synthetic fertilizers. Key GHG like carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) are major drivers of global warming, with rice alone accounting for about 10% of total agri-food system emissions (FAO, 2023). Rice cultivation further produces nearly 50% of cropland CH_4 and around 10% of N_2O emissions (Wang et al., 2020). This highlights the urgent need for sustainable and climate-resilient rice farming practices to reduce emissions and secure future productivity. To address these environmental challenges, several researchers have highlighted the potential of biochar as a promising solution.

Biochar is a carbon-rich material derived from the pyrolysis of agricultural biomass, such as crop residues, wood, and manure, under high temperatures (300–700 °C) in low-oxygen environments (Jeffery et al., 2011; Asirifi et al., 2023). Pyrolysis temperature is a key determinant of biochar characteristics, particularly surface area, porosity, and stability. An increase in temperature from 400°C to 900°C significantly enhances the surface area, from 120 to 460 m^2/g, resulting in biochars with greater adsorptive capacity, similar to activated carbon (Day et al., 2005; Ogawa et al., 2006). While feedstock type strongly influences total carbon, fixed carbon, and ash content, parameters such as volatile matter, nitrogen content, and yield are more sensitive to pyrolysis conditions (Deenik et al., 2010). Biochars porous structure, high surface area, and stable carbon composition improve soil aeration, water retention, nutrient availability, and microbial activity (Adhikari et al., 2022; Abhishek et al., 2022; Zhao et al., 2024). Additionally, its resistance to decomposition contributes to long-term carbon sequestration and reduced greenhouse gas emissions (Weng et al., 2017; Zhang et al., 2023), establishing it as a promising tool for sustainable soil management.

In response to the growing interest around biochars potential to lower emissions and optimize the yield emission trade-offs along with soil health, several studies have been conducted across the world (Liu et al., 2019; Xu et al., 2021; Fang et al., 2024; Wen et al., 2025; Bai et al., 2025). These studies have shown varied results, like a wide range of reduction potential of biochar for CH_4 emissions by 20–91% (Liu et al., 2013). Besides this, mixed effects in case of N_2O emission, with some reporting increases up to 190% and others noting reductions of 28–60% (Liu et al., 2013; He et al., 2017). Several other studies have indicated average reductions in CH_4, N_2O, and global warming potential (GWP) by 5.2%, 14.4%, and 8.6%, respectively (Wen et al., 2025), and a 29% decrease in greenhouse gas intensity (GHGI) (Liu et al., 2019). Biochar application has been reported to reduce CH_4 emissions, likely due to enhanced oxidation processes associated with changes in microbial activity (Pratiwi and Shinogi, 2016), including increased population of methanotrophic bacteria in paddy soils, thereby lowering the ratio of methane-producing to methane-oxidizing microbes (Feng et al., 2012). In terms of N_2O emission, biochar has been shown to decrease emissions by improving soil physical properties, particularly through reduced compaction and lower bulk density (Oomori et al., 2016), which enhances soil aeration and microbial dynamics. Additionally, biochar can interact with reactive nitrogen species by adsorbing nitrate and influencing the transformation of NO_2^- and N_2O, thereby suppressing emissions (Cayuela et al., 2014). However, the effects of biochar are not uniform; some studies have reported increased emissions of CH_4, N_2O, or higher GHGI following application (Singla et al., 2014; Ly et al., 2015; Wang et al., 2021). These inconsistencies are attributed to variations in biochar properties (e.g., feedstock, pyrolysis temperature), soil type, moisture status, crop systems, and microbial community responses. Such variability underscores the need for systematic, context-specific quantification of biochars impacts. In terms of rice productivity, biochar application has been reported to enhance rice yield and contribute to long-term carbon sequestration. Studies have shown rice yield increases of 15.1% with biochar alone and 48.4% when combined with fertilizer (Xu et al., 2021), with other studies reporting yield improvements ranging from 5% to 51% (Biederman and Harpole, 2013; Liu et al., 2013; Cayuela et al., 2014; He et al., 2017; Borchard et al., 2019; Wen et al., 2025). These gains are likely due to enhanced nutrient availability, reduced nutrient leaching, and improved microbial activity (Liu et al., 2013; He et al., 2020). However, several studies have reported mixed responses, with yield increases being inconsistent and effects on other soil properties varying across conditions (Spokas and Reicosky, 2009; Jeffery et al., 2011; Joseph et al., 2010).

So, it is clear that the response of biochar varies widely across regions, soils, and management practices, leading to major uncertainties that limit its large-scale adoption (Spokas and Reicosky, 2009; Jeffery et al., 2011; Ly et al., 2015; Wang et al., 2021). This variability stems from localized studies that often report contrasting results, leaving key gaps in understanding biochars effects on rice yield–emission trade-offs, soil carbon sequestration, and other soil properties. Further uncertainty stems from the lack of standardized application rates, as reported biochar doses range widely across studies. Moreover, a comprehensive understanding remains limited regarding how varied experimental conditions and management practices interact with biochar, such as water regimes (continuous flooding vs. alternate wetting and drying), cultivation methods (puddled transplanted rice vs. direct-seeded rice), and different rates and kinds of input application (biochar and N-fertilizer) affect rice yield-emission trade-offs and TSCSR across diverse agro-environmental contexts. Additionally, biochar characteristics (pH, C:N ratio of feedstock, pyrolysis temperature) and their interactions with soil properties (texture, pH, SOC, TN) may also play critical roles in regulating the yield emission trade-offs and improvement in total soil carbon stock response. Although several earlier meta-analyses (e.g., Liu et al., 2019; Borchard et al., 2019; Wen et al., 2025) have attempted to synthesize biochar impacts, they were constrained by smaller datasets and a narrower range of moderating factors. With the rapid expansion of biochar research in recent years, a substantial number of new studies are now available that allow a more detailed and statistically robust assessment. Building on this expanded evidence base, the present meta-analysis incorporates a significantly larger dataset and applies a more comprehensive categorization of key moderators, including biochar properties, soil characteristics, management practices, and input regimes, far beyond the scope of previous studies. Therefore, a global meta-analysis is essential to elucidate the broader role of biochar in promoting sustainable rice production and enhancing soil carbon sequestration.

To address these critical knowledge gaps, this meta-analysis synthesized 1143 observations from 54 peer-reviewed studies to assess the effects of biochar application on GHG emissions (CH_4, N_2O, GWP), GHGI, TSCSR, and rice grain yield. Specifically, the objectives of this study were to: (1) evaluate the overall impact of biochar application on GHG emissions, TSCSR, rice yield; (2) identify the influence of key moderating factors such as experimental conditions, management practices, soil characteristics, and biochar properties on these outcomes; and, (3) determine the optimal biochar application strategies that simultaneously enhance productivity and reduce environmental impact.

The findings aim to provide data-driven insights for guiding sustainable biochar use in rice production systems and informing future research and policy development.

2. Materials and methods

2.1. Literature search and review process

A systematic review of peer-reviewed literature from 2000 to 2024 was conducted using multiple online databases, following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (Lori et al., 2017) as depicted in Fig. 1. The search employed a range of keywords individually and in combination, including "biochar," "GHG" or "greenhouse gases," "CH_4" or "methane," "N_2O" or "nitrous oxide," "GWP" or "global warming potential," "GHGI" or "greenhouse gas intensity," "TSCSR" or "Total soil carbon stock response," and "rice yield" or "paddy yield." This process initially yielded approximately 27,000 publications. After a thorough screening and refinement process, 54 eligible studies were selected for meta-analysis, comprising 1143 paired observations. These included 197 for methane, 164 for nitrous oxide, 176 for GWP, 176 for GHGI, 336 for grain yield, and 94 for TSCSR, as individual studies often reported multiple treatment comparisons. The geographic distribution of the study sites is illustrated in Fig. 2.

2.2. Screening and criteria of selection

To minimize selection bias, only studies meeting the following criteria were included: (1) data must be derived from field or pot experiments conducted exclusively on rice; (2) studies must include direct comparisons between biochar-treated plots and corresponding control treatments; (3) essential statistical details such as the number of replicates and at least one measure of variability, such as standard deviation, standard error, critical difference, or confidence intervals must be reported; and (4) the study must provide data on at least one of the target variables: N_2O or CH_4 emissions, global warming potential (GWP), greenhouse gas intensity (GHGI), total soil carbon stock response (TSCSR), or grain yield. For each treatment, mean values from replicate measurements were extracted. In cases where a single study reported results from multiple sites or management conditions, each treatment was considered an independent data point. Raw data were obtained directly from tables or, when necessary, digitized from figures using the Web Plot Digitizer tool (version 4.7) (https://automeris.io/).

2.3. Data treatment and groupings

To assess the influence of co-varying factors on CH_4 and N_2O emissions, GWP, GHGI, TSCSR, and grain yield, the dataset was further stratified using four primary categorical moderators: experimental conditions, management practices, soil properties, and biochar characteristics, acknowledging certain limitations in data availability (Table 1). To ensure consistency and transparency in subgroup analysis, all variables were further categorized into predefined classes. Experimental conditions were grouped by experimental design (field and pot), climate (tropical, subtropical, and temperate), and region (China and countries other than China-OTC). Management practices were classified into cultivation methods (PTR and DSR), irrigation regimes (AWD and CF), nitrogen application rates (control: <100 kg ha^{-1}; low: 100–150 kg ha^{-1}; medium: >150–200 kg ha^{-1}; high: >200 kg ha^{-1}), and biochar application rates (control: 0 t ha^{-1}; low: >0–10 t ha^{-1}; medium: >10–30 t ha^{-1}; high: >30 t ha^{-1}). Soil properties were categorized based on texture (coarse, medium, and fine), organic carbon content (low: <0.5%; medium: 0.5–0.75%; high: ≥0.75%), soil pH (acidic: <6.6; neutral: 6.6–7.4; alkaline: >7.4), organic matter content (low: <2%; medium: 2–4%; high: ≥4%), and total nitrogen (low: <100 kg ha^{-1}; medium: 100–150 kg ha^{-1}; high: 150–200 kg ha^{-1}; very high: >200 kg ha^{-1}). Biochar properties were classified according to biochar pH (acidic, neutral, moderately alkaline, and highly alkaline), feedstock C:N ratio (low, medium, and high), and pyrolysis temperature (low: < 650°C; high: ≥650°C).

Global warming potential values were either directly extracted from figures or tables or calculated when CH_4 and N_2O emissions were reported for the same experiment. Soil CO_2 emissions were not included in the analysis due to their high temporal variability and the complexity of interpreting net CO_2 fluxes in flooded rice systems, where simultaneous emission and uptake occur, resulting in context-dependent net fluxes (IPCC, 2013). Consequently, the GWP estimates in this study are based on CH_4 and N_2O emissions, which are the primary contributors to radiative forcing in such systems and are more consistently reported across studies. To determine the overall GWP, CH_4 and N_2O emissions were converted into CO_2-equivalent units using standardized conversion factors. The GWP was calculated according to the guidelines provided by the (IPCC et al., 2007).

$$GWP\ (kg\ CO_2\ \ eq.ha^{-1}) = (CH_4\quad (kg\ ha^{-1}) \times \quad 25\ ) + (N_2O\ (kg\ ha^{-1}) \times \quad 298\ ) \tag{1}$$

where, GWP factors used in this study were 25 for CH_4 and 298 for N_2O, based on a 100-year time horizon (Reddy et al., 2025). Emission factors for agricultural inputs were derived from the Eco invent Database v3.9 & v3.9.1 (Moreno Ruiz et al., 2020) and the Chinese Life Cycle Database (CLCD) (Liu et al., 2020).

Greenhouse gas intensity was calculated as the ratio of total GWP to rice yield for each treatment, following the method outlined by Zhang et al. (2011) and Shang et al. (2011).

$$GHGI\ (\text{kg}\ \ CO_2 - \text{eq.}\ \ \text{kg}^{-1}\text{grain}) = \frac{GWP}{Rice\ grain\ yield} \tag{2}$$

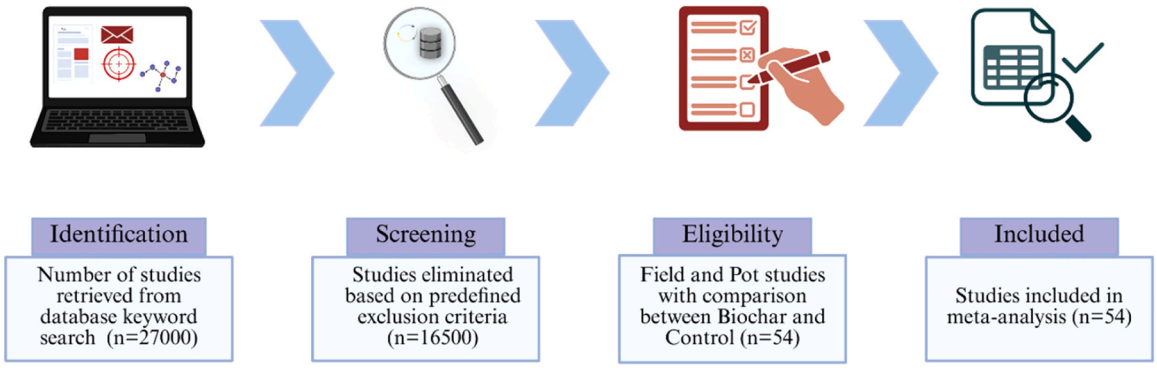


Fig. 1. The chart provides an overview of the identified, excluded, and included studies in this meta-analysis. PRISMA flow diagram modified from Lori et al. (2017).

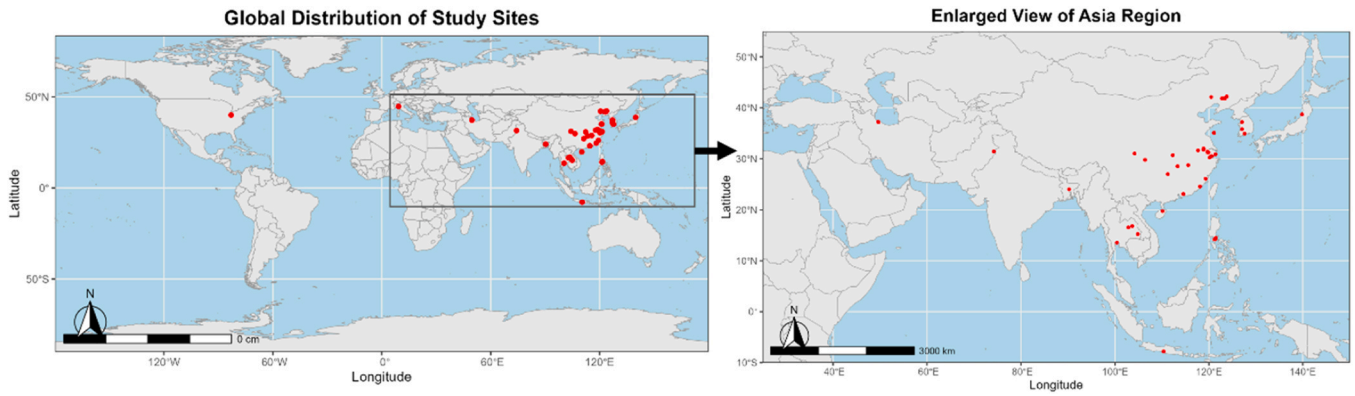


Fig. 2. Geographical distribution of studies.

Table 1
Categorization of factors for meta-analysis (CH_4, N_2O, GWP, GHGI, TSCSR, and Grain Yield).

Experimental Conditions	Experimental design	Field, Pot
	Climate	Tropical, Subtropical, Temperate
	Region	China, OTC (countries other than China)
Management Practices	Cultivation method	PTR, DSR
	Irrigation method	AWD, CF
	Nitrogen application rate (kg/ha)	Control (<100), Low (≥ 100 and ≤ 150), Medium (> 150 and ≤ 200), High (> 200)
	Biochar application rate (t/ha)	Control (0), Low (> 0 and ≤ 10), Medium (> 10 and ≤ 30), High (> 30)
Soil Properties	Texture	Coarse, Medium, Fine
	Organic carbon (%)	Low (<0.5), Medium (≥ 0.5 and < 0.75), High (≥0.75)
	pH	Acid (<6.6), Neutral (6.6–7.4), Alkaline (>7.4)
	Organic matter (%)	Low (<2), Medium (≥ 2 and < 4), High (≥4)
	Total nitrogen (kg/ha)	Low (<100), Medium (100–150), High (150–200), Very High (200)
Biochar Properties	pH of biochar	Acidic, Neutral, Moderately Alkaline, Highly Alkaline
	C:N ratio of feedstock	Low C:N ratio, Medium C:N ratio, High C:N ratio
	Pyrolysis temperature (°C)	Low (< 650°C), High (≥650°C)

2.3.1. Harmonization of total soil carbon stock response

In this meta-analysis, TSCSR is defined as the annualized change in SOC stocks, expressed in t C ha^{-1} yr^{-1} . When primary studies reported SOC as concentrations (% C), values were converted to stock changes using soil bulk density (BD, g cm^{-3}) and sampling depth (D, cm) as follows:

$$SOC\ stock\ change\ (t\ C\ ha^{-1}) = \Delta SOC\ (\%) \times BD\ (g\ cm^{-3}) \times D\ (cm) \quad (3)$$

To account for differences in experimental duration among studies, SOC stock changes were annualized by dividing the total SOC stock change by the corresponding study duration (years) to obtain TSCSR. Estimated SOC stock values were included only when they could be consistently expressed in the same units. For studies using different definitions of SOC change, we conducted sensitivity checks comparing results including or excluding these studies, and the overall patterns of TSCSR remained consistent. This procedure ensures comparability and transparency of the aggregated TSCSR values across diverse primary studies.

It is important to note that the calculated TSCSR values represent changes in total soil carbon following biochar application and may include contributions from both biochar-derived carbon and native SOC. Due to limitations in the primary studies, it was not possible to distinguish among these carbon pools (e.g., using isotopic tracing or carbon fractionation). Therefore, the reported TSCSR values should be interpreted as total carbon responses rather than exclusive estimates of native SOC sequestration.

2.4. Effect size derivation and statistical analysis

The effect size used in this study was the response ratio (RR), where the biochar treatment (E) was considered the experimental variable and compared against the control (C), representing treatments without biochar. The response ratio was calculated following the approach described by Hedges et al. (1999):

$$RR = \frac{X_E}{X_C} \quad (4)$$

where, X_E and X_C represent the conveyed mean values for the biochar-treated and control treatments, respectively, for each observation.

$$\ln RR = \ln\frac{X_E}{X_C} = \ln(X_E) - \ln(X_C) \quad (5)$$

The natural logarithm of each response ratio (lnRR) was calculated to normalize the data distribution across studies. An lnRR of zero indicates no effect of the treatment, while values less than or greater than zero suggest negative or positive effects, respectively. The sampling variance (*vi*) associated with each lnRR was determined using the equation provided by Hedges et al. (1999):

$$vi = \frac{(SD_E)^2}{n_E\, x\, (X_E)^2} + \frac{(SD_C)^2}{n_C\, x\, (X_C)^2} \quad (6)$$

where, X represents the mean, SD the standard deviation, and n the sample size for both Biochar and Control treatments in each case.

When standard deviations were not reported directly, other measures of variability such as the coefficient of variation (CV) and least significant difference (LSD) were used to estimate them. If CV was provided for a given response variable, it was multiplied by the corresponding mean to derive the SD. In cases where LSD was reported, it was converted into standard error (SE) using the formula described by Quinn et al. (2020), as shown below.

$$SE = \frac{LSD}{t_{0.975,\ n}\sqrt{2bn}} \quad (7)$$

where, t denotes the *t*-test value, n represents the number of samples, and b indicates the number of replications. To calculate the SD, the derived SE was multiplied by the square root of the sample size (n), as shown in the equation below:

$$SD = SE\ x\ \sqrt{n} \tag{8}$$

where, n refers to the sample size. Additionally, when the CI and mean were reported, the SD was calculated using the following formula:

$$SD = (CI_u - CI_l)\ x \frac{\sqrt{n}}{2Z_{\frac{\alpha}{2}}} \tag{9}$$

where, CI_u and CI_l represent the upper and lower bounds of the confidence interval, respectively, while $Z_{\alpha/2}$ denotes the Z-score corresponding to the chosen significance level, 1.645 for $\alpha = 0.10$, and 1.96 for $\alpha = 0.05$. In cases where neither SD, CI, nor SE were reported, the standard deviation was approximated as one-tenth of the mean, following the approach used by Tian et al. (2015).

The weighting factor (wi) for each study was calculated as the inverse of the total variance under a random-effects model: $wi = 1 / (vi + \tau^2)$, where vi is the within-study variance and τ^2 represents the between-study variance. The mean effect sizes of response ratios (RRs) were then estimated through a weighted meta-analysis using the following formula:

$$\overline{\ln RR} = \frac{\sum (\ln RRi x wi)}{\sum wi} \tag{10}$$

where, $\ln RRi$ represents the effect size, and wi denotes the weight assigned to the ith comparison. To facilitate interpretation, all results were expressed as percentage changes across moderators, calculated using the following formula:

$$Effect\ Size\ (\%) = (\exp(\overline{\ln RR}) - 1) \times 100 \tag{11}$$

We employed a random-effects model to calculate grouped effect sizes using the "metafor" package in R (version 4.2.3; R Core Team, 2022). This model was selected due to the wide geographic distribution of the included studies, which reflect diverse environmental, agricultural, and management conditions. Unlike the fixed-effects model which assumes a single true effect size across all studies, the random-effects model accounts for both within-study variation and between-study heterogeneity, allowing effect sizes to vary due to differences in location, climate, soil types, and crop practices. Given the heterogeneity in our dataset, the random-effects model offers a more realistic and generalized understanding of overall trends by accommodating study-specific variability. This approach ensures a more robust and accurate synthesis of findings drawn from varied regional contexts.

Effect sizes were calculated as log response ratios (lnRR) and analysed using a random-effects model (REML). A categorical variable was considered to have a significant effect when the 95% confidence interval (CI) of its mean effect size did not include zero. Differences among subgroup levels were evaluated using corresponding p-values from subgroup analyses or meta-regression, rather than relying on overlap of 95% CIs. Where statistical support was lacking, differences were described as numerical trends. Heterogeneity among studies was assessed using Cochran's Q, I^2, and τ^2 statistics. Publication bias was evaluated using Egger's regression test and the trim-and-fill method. Dose–response relationships and moderator effects were examined using meta-regression analyses. All statistical analyses and visualizations, including forest plots, funnel plots, and regression analyses of response ratios (RR), were conducted using R software (R Core Team, 2022). Detailed statistical outputs, including model estimates, heterogeneity statistics, publication bias tests, and meta-regression results, are provided in the Supplementary Tables (S1-S37).

3. Results

3.1. Effect of biochar and non-biochar treatments on GHG emissions, GHGI, TSCSR, and grain yield

The meta-analysis revealed statistically significant effects across all evaluated variables, including, CH_4 (n = 197), N_2O (n = 164), GWP (n = 176), GHGI (n = 176), TSCSR (n = 94), and grain yield (n = 362). The percentage changes reported here are based on back-transformed log response ratio (lnRR) values, and details are provided in the **Supplementary file** (Table S14**- S37**). Across emission-related variables, CH_4, N_2O, GWP, and GHGI showed significant reductions of 18.11%, 15.54%, 25.51%, and 33.16%, respectively. Whereas, TSCSR and grain yield increased by 69.96% and 11.13%, respectively **(**Fig. 3**)**. Funnel plots were generated separately for each study variable to assess the distribution of effect sizes, study precision, and potential small-study effects or publication bias (Fig. 4). These plots display the observed effect sizes (*lnRR*) on the x-axis and their corresponding standard errors (a measure of precision) on the y-axis. Similarly, Supplementary Figure 1 presents the distribution of *lnRR* values for each outcome variable under various studies, with all Figs. (1–4) provided in **Supplementary File**.

The heterogeneity statistics for all outcome variables, including Cochran's Q, I^2, and τ^2, were calculated using random-effects meta-analysis. The results indicated substantial between-study variability across all variables. Across all variables, heterogeneity was consistently high ($I^2 > 90\%$ for CH_4, N_2O, GWP, GHGI, TSCSR, and yield), indicating substantial variation in effect sizes beyond sampling error, likely driven by differences in environmental conditions, biochar properties, and experimental designs. Detailed statistics are provided in Supplementary Tables S1–S6.

3.2. Effect of experimental condition on GHG emissions, GHGI, TSCSR, and grain yield

3.2.1. Comparative effect of biochar and non-biochar treatments on GHG emissions and GHGI

The effects of biochar on GHG emissions and related indices varied across experimental conditions, with contrasting patterns observed between field and pot studies **(**Fig. 5**)**. Under field conditions, biochar application was associated with significant reductions in CH_4, N_2O, GWP, and GHGI, with average declines of 24.57%, 18.89%, 31.14%, and 37.08%, respectively. In contrast, pot experiments showed no clear mitigation pattern, with CH_4 and GWP tending to increase (19.8% and 17.03%, respectively), while N_2O and GHGI showed only marginal changes.

Climatic conditions further influenced the magnitude of these responses. In subtropical regions, significant reductions were observed across all variables, with CH_4, N_2O, GWP, and GHGI decreasing by 19.57%, 12.99%, 17.56%, and 27.46%, respectively. In temperate regions, the response was more pronounced for N_2O, GWP, and GHGI, which significantly declined by 32.96%, 36.69%, and 36.20%, respectively. Tropical regions showed the significant reductions in integrated indices, with GWP and GHGI decreasing by 44.30% and 48.73%, respectively.

A similar pattern was evident across geographical regions. In China, biochar application was associated with significant reductions in CH_4, N_2O, GWP, and GHGI by 18.9%, 13.19%, 25.74%, and 31.20%, respectively. Studies conducted in OTC (countries other than China) also showed significant reductions, with CH_4, N_2O, GWP, and GHGI declining by 17.04%, 26.91%, 24.17%, and 45.36%, respectively. However, this OTC category should be interpreted with caution, as it represents an aggregated group of diverse agroecosystems and does not allow for country-specific inference. Overall, these findings indicate that the mitigation potential of biochar varies with experimental conditions, climate, and regional factors.

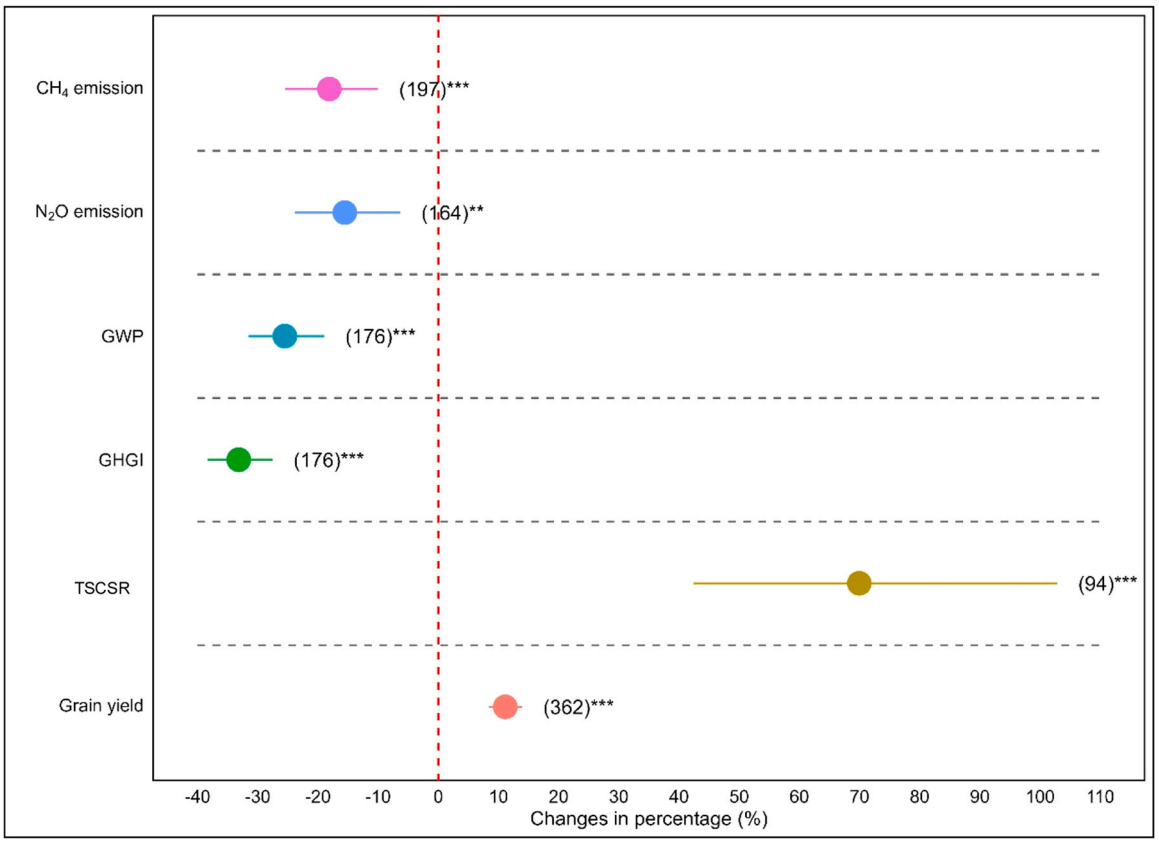


Fig. 3. Overall effects of Biochar on greenhouse gas (CH_4 and N_2O) fluxes, global warming potential (GWP), greenhouse gas intensity (GHGI), total soil carbon stock response (TSCSR), and grain yield. Parentheses numbers indicate the number of observations, and error bars represent 95% confidence intervals; at *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$.

3.2.2. Comparative effect of biochar and non-biochar treatments on TSCSR

Biochar application was associated with increases in TSCSR across experimental conditions, climates, and regions. The TSCSR was significantly higher under field conditions (71.75%) than in pot experiments (35.41%). Across climatic zones, the response varied considerably, with the significant increase observed in subtropical regions (196.95%), followed by tropical (35.67%) and temperate regions (35.34%). A similar trend was evident across geographical regions, where studies conducted in China showed a significant increase (160.38%) compared to OTC (39.81%).

3.2.3. Comparative effect of biochar and non-biochar treatments on grain yield

Biochar application was associated with improvements in grain yield under both experimental conditions. Yield increases were significantly higher in pot studies (21.96%) than in field conditions (9.85%). Climatic variation influenced yield response, with significant increases observed in subtropical (15.25%) and tropical regions (4.93%), while only marginal changes were observed in temperate regions (1.90%). Regionally, grain yield increased in both China (10.18%) and OTC (12.23%), indicating a significant positive response across locations.

3.3. Effect of management practices on GHG emissions, GHGI, TSCSR, and crop yield

3.3.1. Comparative effect of biochar and non-biochar treatments on GHG emissions and GHGI

Management practices influenced the magnitude of biochar-induced changes in GHG emissions and GHGI (Fig. 6). Overall, biochar application was associated with reductions in emission-related variables across most management conditions, although the extent of response varied. Under cultivation methods, PTR showed significant reductions in CH_4, GWP, and GHGI by 19.62%, 26.01%, and 33.51%, respectively. In contrast, DSR showed significant reductions in GWP (10.69%) and GHGI (23.12%), while CH_4 tended to increase (16.28%). N_2O emissions declined under both systems, with a significantly higher reduction observed in DSR (39.47%) compared to PTR (14.43%).

Irrigation practices also affected emission responses. AWD was associated with significant reductions in N_2O, GWP, and GHGI by 20.90%, 30.83%, and 35.07%, respectively. Under controlled flooding (CF), decreases were observed across CH_4, N_2O, GWP, and GHGI, with significant reductions of 23.15%, 8.78%, 22.05%, and 29.87%, respectively.

Nitrogen application rates further modulated these effects. The significant reductions in CH_4, GWP, and GHGI were observed at low and medium nitrogen levels, with CH_4 decreasing by 30.60% and 25.73%, respectively. GWP significantly declined by 21.48% (low) and 41.00% (medium), while GHGI showed corresponding significant reductions of 41.16% and 44.29%. N_2O emissions showed a significant decline at low nitrogen levels (67.59%), whereas high nitrogen levels resulted in a slight increase.

Biochar application rate also played a key role in determining outcomes. Medium application rates resulted in the significant reductions in N_2O (40.07%), GWP (38.36%), and GHGI (45.56%). Low rates also significantly reduced GWP (23.71%) and GHGI (31.09%), while CH_4 decreased significantly under both low (26.16%) and medium (24.74%) application rates. Overall, moderate biochar application rates tended to provide more consistent mitigation benefits.

3.3.2. Comparative effect of biochar and non-biochar treatments on TSCSR

Biochar application was associated with increases in TSCSR across all management practices, with the magnitude of response varying by

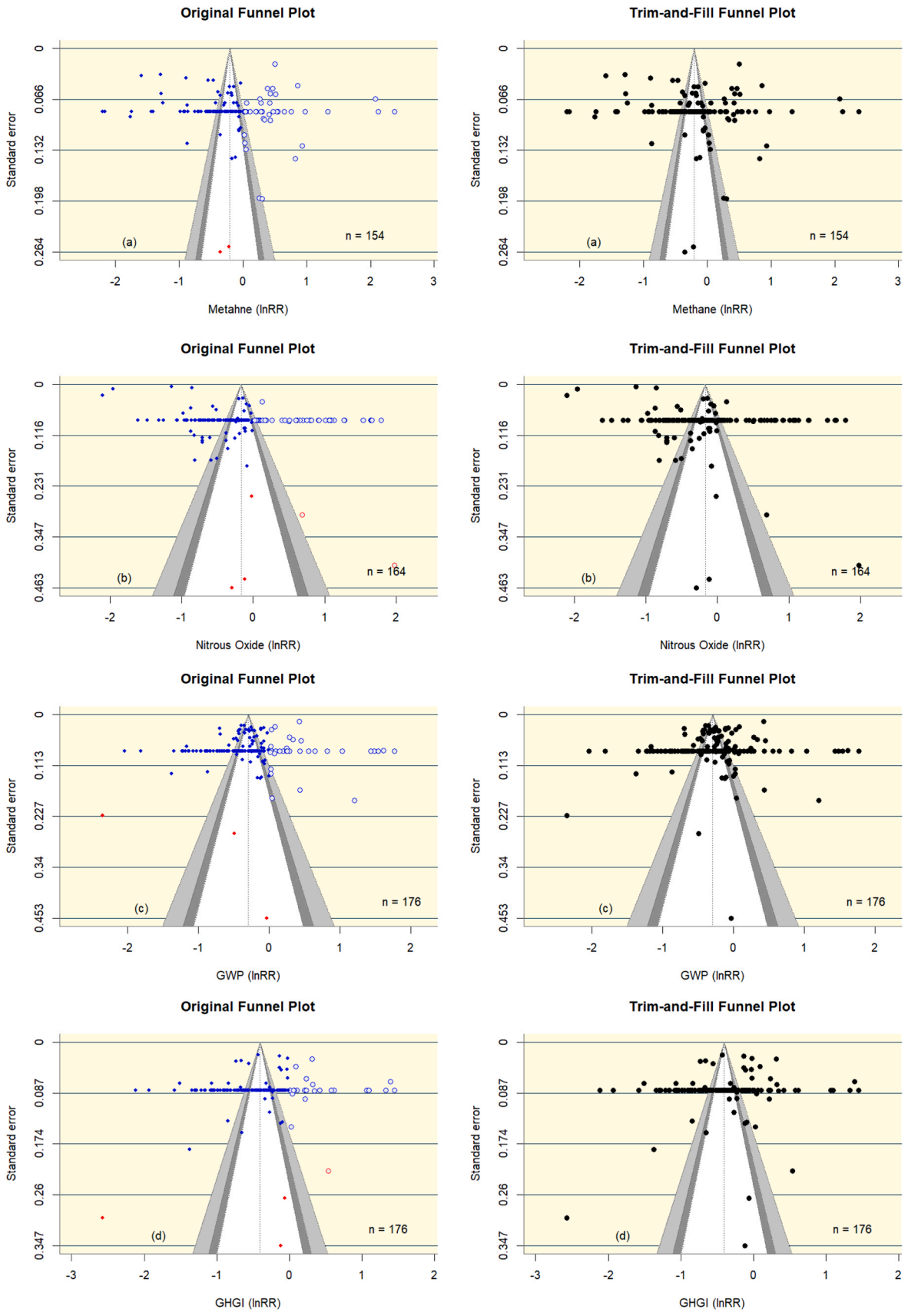


Fig. 4. The funnel plots along with Trim-and-fill funnel plot depict the precision of individual studies for (a) CH_4, (b) N_2O, (c) GWP, (d) GHGI, (e) TSCSR, and (f) Grain yield. Studies with low (< 0.2 SE) and high (> 0.2 SE) standard errors are represented by blue and red dots, respectively. Filled circles denote studies showing a positive lnRR compared to the overall mean, while unfilled circles indicate a negative response. (For interpretation of the colour references in this figure legend, please refer to the online version of the article.).

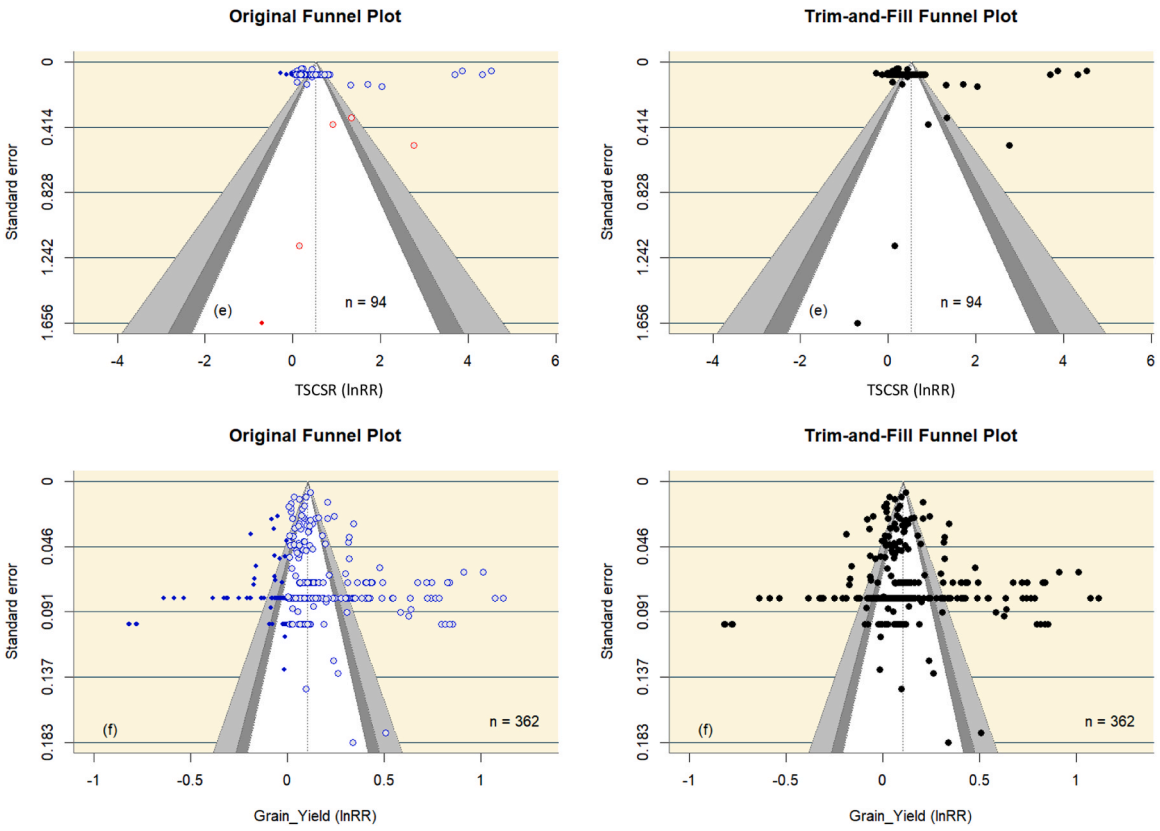


Fig. 4. (*continued*).

cultivation method, irrigation, nutrient level, and application rate. Among cultivation methods, significant increases were observed under PTR (81.25%) compared to DSR (32.9%). Similarly, AWD resulted in significantly higher increases (108.18%) than CF (50.02%). Nitrogen levels showed a significant increase in TSCSR, with values rising from 17.69% under control to 44.3%, 114.35%, and 140.29% under low, medium, and high nitrogen rates, respectively. Biochar application rates exhibited a significant peak response at medium levels (117.71%), followed by high (73.48%) and low rates (43.03%). Overall, larger increases in TSCSR were associated with PTR, AWD, and moderate to high nitrogen and biochar application rates.

3.3.3. Comparative effect of biochar and non-biochar treatments on grain yield

Biochar application was associated with improvements in grain yield across management practices, although the magnitude of response varied. Yield increases were significantly higher under PTR (11.57%) compared to DSR (6.48%). Similarly, AWD resulted in significant improvements (12.64%) than CF (8.71%). Nitrogen availability strongly influenced yield response, with a decline observed under control conditions (-14.21%), while increases were recorded under low (9.88%), medium (8.8%), and high nitrogen rates (19.21%). Biochar application rate showed a gradual increase in yield from control (6.99%) to low (13.58%) and medium rates (14.37%), followed by a decline at higher application levels (6.05%). Overall, stronger yield responses were associated with PTR, AWD, and low to medium nitrogen and biochar application rates.

3.4. Effect of soil properties on GHG emissions, GHGI, TSCSR, and grain yield

3.4.1. Comparative effect of biochar and non-biochar treatments on GHG emissions and GHGI

Soil properties influenced the response of GHG emissions and GHGI to biochar application (Fig. 7), with variations observed across soil texture, organic carbon (OC), organic matter (OM), pH, and total nitrogen (TN).

Across soil textures, reductions in CH_4 and GWP were more pronounced in coarse and medium-textured soils, with CH_4 significantly decreased by 32.12% and 21.00%, respectively. N_2O declined significantly in fine (43.52%) and coarse soils (33.66%), while GHGI significantly decreased across all soil textures, with the largest reduction observed in coarse soils (49.18%).

Soil organic carbon levels further modified these responses. The significant reductions in CH_4 (28.50%), N_2O (42.72%), GWP (29.15%), and GHGI (37.34%) were observed under medium OC conditions, while comparatively smaller reductions occurred under low and high OC soils.

A similar pattern was evident for soil organic matter. CH_4 significantly decreased under low (24.70%) and medium OM (17.53%) levels but increased under high OM conditions (30.77%). In contrast, N_2O, GWP, and GHGI declined significantly across all OM levels. The significant reductions in GWP ranging from 13.55% (high OM) to 28.07% (medium OM) and GHGI from 17.47% (high OM) to 34.51% (medium OM).

Soil pH also influenced emission responses, with CH_4 reductions significantly increasing from acidic (16.70%) to alkaline soils (33.14%). GWP and GHGI showed significant declines across pH levels, with reductions of up to 40.31% and 39.62%, respectively, under alkaline conditions. N_2O reductions were more pronounced under neutral soils (37.61%).

Total nitrogen further modulated biochar effects. CH_4 reductions

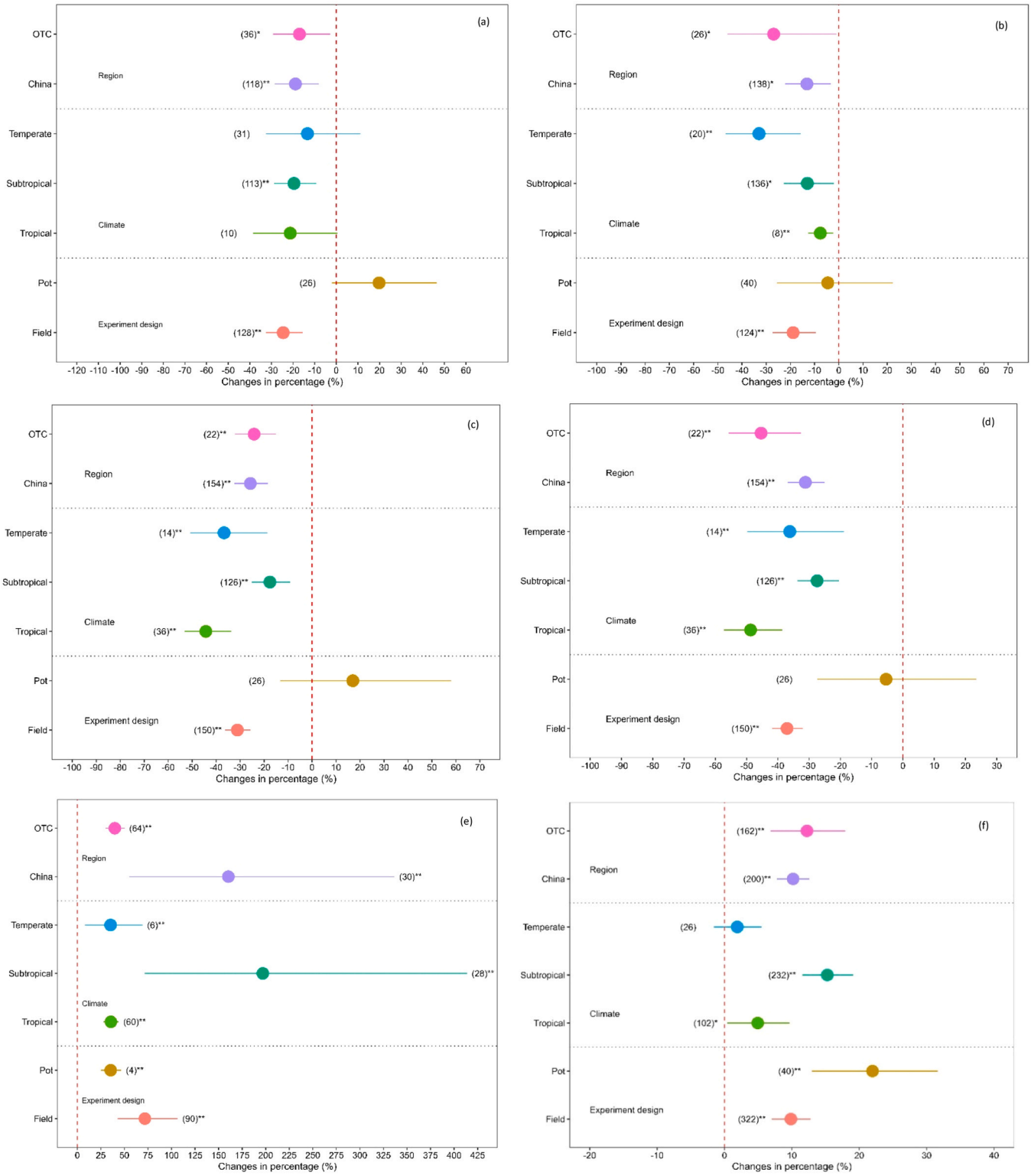


Fig. 5. Effects of Experimental conditions on CH_4 and N_2O fluxes, global warming potential (GWP), Greenhouse gas intensity (GHGI), total soil carbon stock response (TSCSR), and grain yield (a-f respectively). Parentheses numbers indicate the number of observations, and error bars represent 95% confidence intervals; at *p < 0.05, **p < 0.01, and ***p < 0.001. OTC denotes countries other than China.

were more evident under low (23.75%) and medium TN levels (25.70%), while N_2O declined significantly under low (39.43%) and high TN conditions (29.29%). GWP and GHGI significantly decreased across all TN levels, with GHGI reductions reaching up to 37.48% (high TN).

3.4.2. *Comparative effect of biochar and non-biochar treatments on TSCSR*

Biochar application was associated with increases in TSCSR across all soil properties, although the magnitude of response varied. Across soil textures, significantly greater increases were observed in medium (72.73%) and fine-textured soils (26.09%). SOC levels also influenced TSCSR, with significant increases observed under low (21.42%),

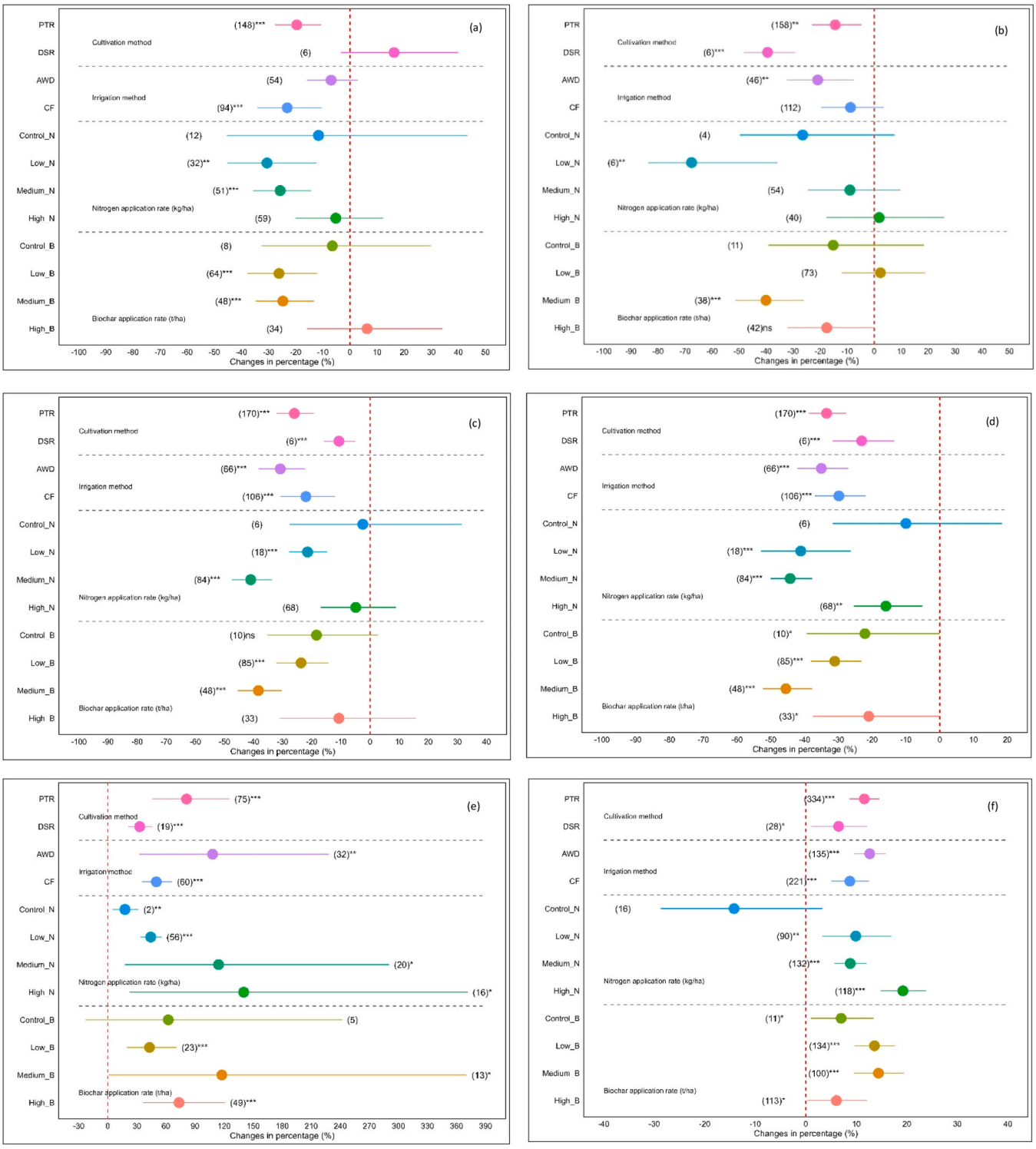


Fig. 6. Effects of Management practices on greenhouse gas (CH_4 and N_2O) fluxes, global warming potential (GWP), Greenhouse gas intensity (GHGI), total soil carbon stock response (TSCSR), and grain yield (a-f respectively). Parentheses numbers indicate the number of observations, and error bars represent 95% confidence intervals; at *p < 0.05, **p < 0.01, and ***p < 0.001.

medium (13.24%), and high SOC conditions (76.44%). A similar pattern was evident across soil organic matter levels, where TSCSR increased significantly under low (30.41%), medium (86.45%), and high OM soils (32.24%). Soil pH further influenced the response, with significantly higher increases under acidic soils (74.72%) compared to neutral conditions (21.29%). Across TN levels, TSCSR increased significantly, with values of 57.66%, 31.16%, 81.96%, and 89.12% under low, medium, high, and very high TN levels, respectively.

3.4.3. Comparative effect of biochar and non-biochar treatments on grain yield

Biochar application was associated with improvements in grain yield across a range of soil properties. Yield responses varied significantly with soil texture, with the highest increases observed in fine-textured



Fig. 7. Effect of Soil properties on greenhouse gas (CH_4 and N_2O) fluxes, global warming potential (GWP), Greenhouse gas intensity (GHGI), total soil carbon stock response (TSCSR), and grain yield (a-f respectively). Parentheses numbers indicate the number of observations, and error bars represent 95% confidence intervals; at *p < 0.05, **p < 0.01, and ***p < 0.001.

soils (31.27%), followed by medium (10.55%) and coarse soils (6.35%). Soil organic carbon showed a relatively consistent positive response, with significant yield increases of 10.95%, 11.94%, and 9.64% observed under low, medium, and high SOC levels, respectively. Across soil organic matter levels, yield significantly increased under low (16.55%), medium (7.84%), and high OM conditions (6.94%). Similarly, significantly higher increases were observed in acidic soils (14.21%) compared to neutral soils (8.02%). Total nitrogen also influenced yield response, with significantly greater increases under low (20.57%) and medium (9.02%) TN levels compared to high TN (6.14%).

3.5. Effect of biochar properties on GHG emissions, GHGI, TSCSR, and grain yield

3.5.1. Comparative effect of biochar and non-biochar treatments on GHG emissions and GHGI

Biochar properties, including pH, feedstock C:N ratio, and pyrolysis temperature, influenced the magnitude of changes in GHG emissions and GHGI (Fig. 8). Biochar pH showed a clear effect on emission responses. Acidic biochar was associated with the significant reductions in CH_4, GWP, and GHGI (38.06%, 37.30%, and 55.59%, respectively). With increasing pH, the magnitude of reduction generally declined, with moderate decreases observed under neutral and moderately alkaline biochar, and relatively smaller changes under highly alkaline conditions. N_2O emissions also declined across pH levels, with stronger

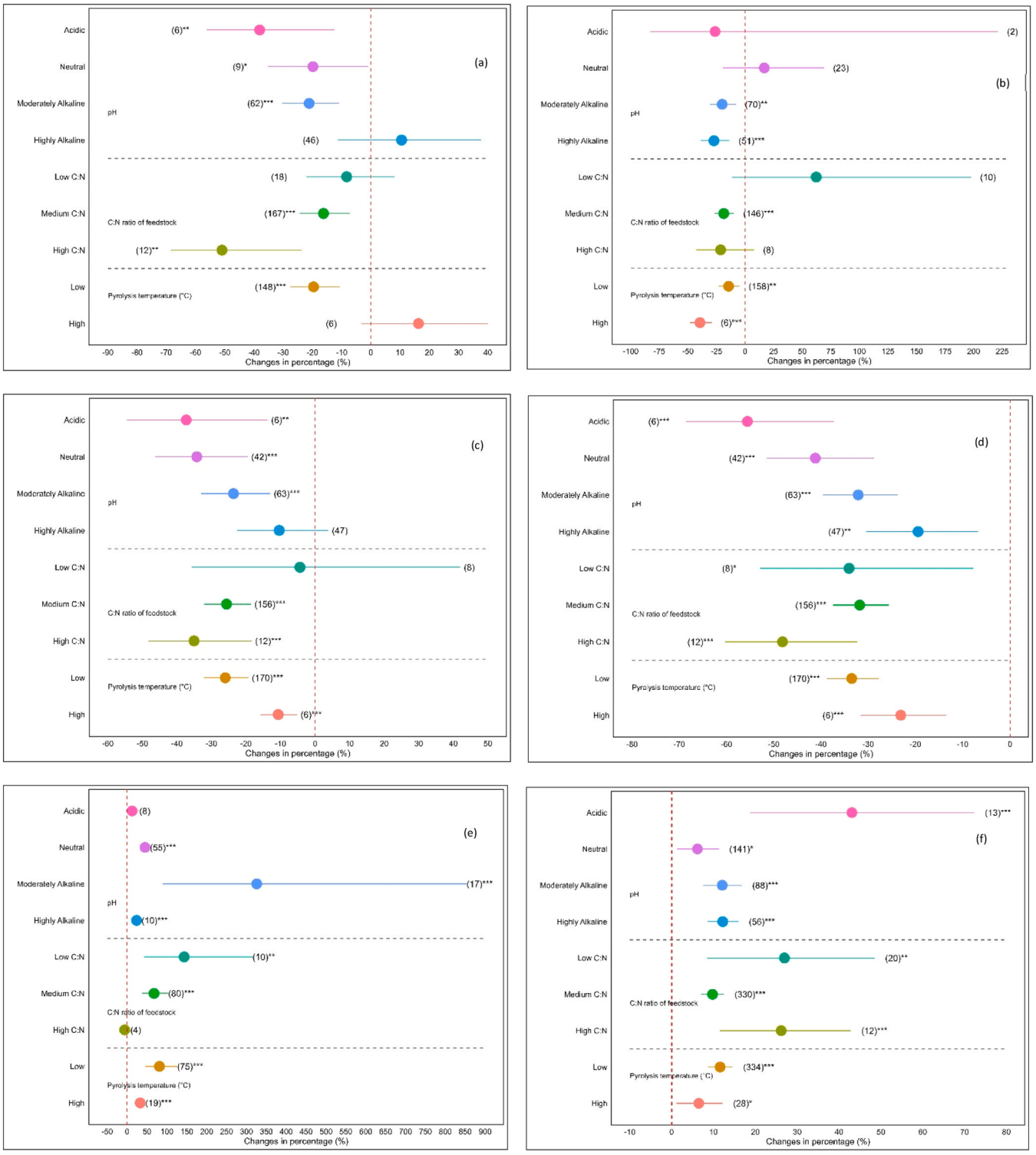


Fig. 8. Effect of Biochar Properties on greenhouse gas (CH_4 and N_2O) fluxes, global warming potential (GWP), Greenhouse gas intensity (GHGI), total soil carbon stock response (TSCSR), and grain yield (a-f respectively). Parentheses numbers indicate the number of observations, and error bars represent 95% confidence intervals; at *p < 0.05, **p < 0.01, and ***p < 0.001.

reductions under moderately to highly alkaline biochar.

Feedstock characteristics further influenced mitigation potential. Biochar derived from high C:N ratio feedstocks showed the significantly largest reductions in CH_4, GWP, and GHGI (50.94%, 35.05%, and 48.15%, respectively). Medium C:N ratio feedstocks also resulted in significant reductions across CH_4, N_2O, GWP, and GHGI, whereas low C:N ratio biochar showed comparatively smaller or inconsistent changes. The pyrolysis temperature exhibited contrasting effects. Low-temperature biochar was associated with significant reductions in CH_4, GWP, and GHGI (19.62%, 26.01%, and 33.51%, respectively), while high-temperature biochar showed significantly stronger reductions in N_2O (39.47%) but relatively smaller effects on GWP and GHGI.

3.5.2. Comparative effect of biochar and non-biochar treatments on TSCSR

Biochar properties influenced the magnitude of changes in TSCSR, with considerable variation across pH, feedstock type, and pyrolysis temperature. Across biochar pH levels, significant increases in TSCSR were observed under neutral (45.05%), and highly alkaline conditions (23.77%), with the largest response under moderately alkaline biochar. Feedstock characteristics also played an important role. Biochar derived from low C:N ratio feedstocks showed the significantly highest increase in TSCSR (143.44%), followed by medium C:N ratio feedstocks (67.59%). Pyrolysis temperature influenced the response, with low-temperature biochar associated with a significantly larger increase in TSCSR (81.25%) compared to high-temperature biochar (32.90%).

3.5.3. Comparative effect of biochar and non-biochar treatments on grain yield

Biochar properties were consistently associated with improvements in grain yield, although the magnitude of response varied across categories. Yield responses differed across biochar pH levels, with the significantly largest increase observed under acidic biochar (43.06%), followed by smaller but significant increases under neutral (6.15%), moderately alkaline (12.07%), and highly alkaline conditions (12.19%). Feedstock type further influenced yield, with the significant improvement under low C:N ratio biochar (26.94%), followed by high (26.16%) and medium C:N ratio feedstocks (9.75%). Pyrolysis temperature showed a similar pattern, with significantly higher yield increases under low-temperature biochar (11.57%) compared to high-temperature biochar (6.48%).

3.6. Relationships between GHGI, TSCSR, grain yield under different biochar application rates

The inter-relationships were established among the lnRR responses of GHG emission, GHGI, and grain yield with biochar application rates, resulting in a non-significant effect as depicted in supplementary fig.2. The lnRR response of GHG with grain yield also reported a non-significant effect (supplementary fig.3). A quadratic relationship between GHGI and biochar application rates was observed (Fig. 9**A**), with GHGI tending to decrease with increasing biochar application, indicating potential mitigation of emissions intensity. Likewise, with increasing biochar rate, TSCSR increased, reflecting enhanced total soil carbon stock (Fig. 9**B**). The quadratic relationship between grain yield and biochar application rate (Fig. 9**C**) was significant ($p < 0.001$), suggesting a non-linear response of yield to increasing biochar inputs.

3.7. Dose-response meta-regression

To better quantify the influence of biochar application rate as a continuous variable, a meta-regression analysis was conducted using biochar application rate (t ha^{-1}) as a continuous moderator to evaluate its relationship with total soil carbon stock response. The analysis revealed a significant positive relationship between biochar application rate and TSCSR, indicating that increasing biochar inputs generally enhanced soil carbon accumulation. The fitted linear model (**lnRR = a + b × Rate**) showed a positive and statistically significant slope ($p < 0.05$), suggesting that higher biochar application rates were associated with greater total soil carbon stock responses (Fig. 10).

To assess potential non-linear responses, a quadratic meta-regression including the squared term ($rate^2$) was also tested. However, the quadratic term was not statistically significant ($p > 0.05$), indicating no clear evidence of a saturation or curvature in the dose-response relationship within the range of application rates evaluated. This indicates that any apparent plateau or reduced response at higher rates observed in subgroup analyses should be interpreted cautiously, as it is not supported by the meta-regression results. Model comparison using the Akaike Information Criterion (AIC) further supported the linear model. These results suggest that TSCSR generally increases with increasing biochar application rate, which may be attributed to the recalcitrant nature of biochar carbon and its ability to stabilize soil organic matter. Although the meta-regression did not identify a statistically defined optimum dose, subgroup trends indicated that moderate biochar application rates (approximately 10–30 t ha^{-1}) frequently resulted in substantial increases in total soil carbon stock response. These findings complement the subgroup analysis, indicating that observed category-based responses reflect a broader continuous dose-dependent relationship rather than discrete threshold effects.

4. Discussion

4.1. Effect of biochar on GHG emissions, GHGI, TSCSR, and grain yield

Overall, the present meta-analysis identified consistent trends that help address previously reported knowledge gaps in rice-based systems. Specifically, the analysis indicates that biochar application was associated with reductions in major GHG emissions, with CH_4 and N_2O emissions declining by 18.11% and 15.54%, respectively. These reductions collectively contributed to a 25.51% decrease in the overall GWP. In addition to its mitigation benefits, biochar application also improved soil health and crop productivity in rice systems, as reflected by a 69.96% increase in TSCSR and an 11.13% enhancement in grain yield. These improvements ultimately resulted in a 33.16% reduction in GHGI. The reduction in GHGI was primarily associated with increased grain yield, while changes in GHG emissions contributed to a lesser extent. However, it is important to interpret the reported increase in TSCSR with caution. The observed increase reflects changes in total soil carbon following biochar application and likely includes both direct inputs of biochar-derived carbon and potential changes in native SOC. Because most primary studies did not separate these components (e.g., via isotopic tracing or carbon fractionation methods), the reported percentage increase may overestimate the contribution of native SOC sequestration. Therefore, these values should not be interpreted as purely representing intrinsic SOC accumulation but rather as an integrated measure of total carbon dynamics in biochar-amended systems. By quantifying these effects across a wide range of studies and conditions, this analysis directly addresses the previously noted inconsistencies and underscores biochars potential as a climate-smart soil amendment. These findings guide the use of biochar in sustainable rice-based systems. Similar findings were also documented by Liu et al. (2019) where they reported that the biochar application significantly reduced GWP by 21% and GHGI by 29% while yield increased by 11%. Moreover, Xu et al. (2021) reported that soil organic carbon increased by 33.5% with biochar application. These findings highlight the crucial role of biochar in mitigating GHG emissions, enhancing the TSCSR, and improving grain yield. Several mechanisms have been proposed to explain how biochar contributes to these positive effects. However, these mechanisms are not universally dominant and may vary depending on environmental conditions, soil properties, and management practices across studies. When applied to paddy soil, biochar improves soil aeration, which promotes CH_4 oxidation and/or suppresses its

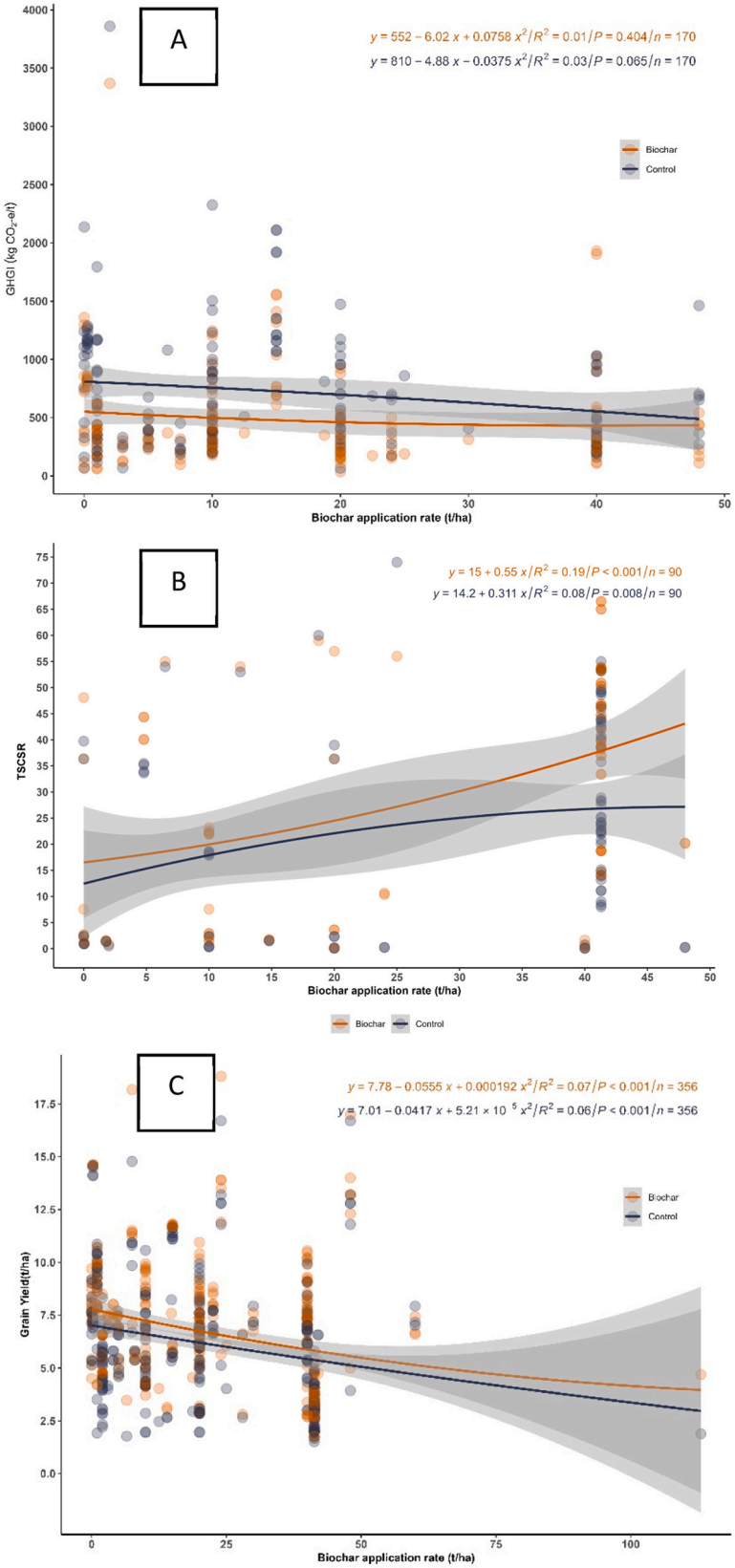


Fig. 9. A-C. Relationship between the GHGI, TSCSR, and grain yield and biochar application rate under both biochar and control conditions.

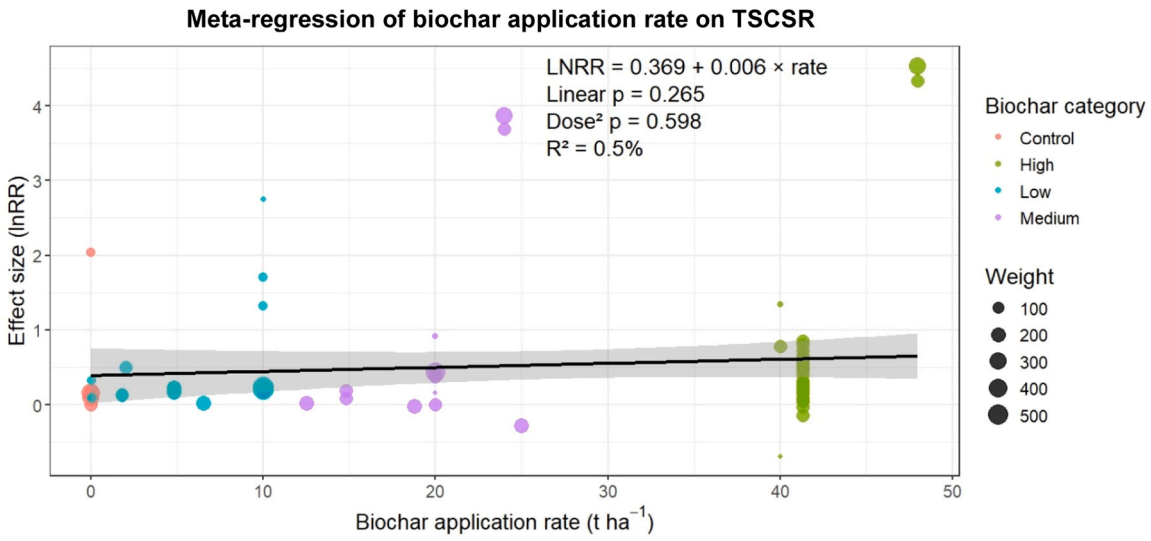


Fig. 10. Dose response meta regression between biochar application rate and total soil carbon stock response.

production (Jeffery et al., 2016). Additionally, biochar has a high surface area that facilitates CH_4 adsorption, thereby reducing its release into the atmosphere (Yaghoubi et al., 2014). It also fosters an increase in methanotrophic microbial populations, which actively consume CH_4 (Feng et al., 2012). Another key mechanism involves the inhibition of CH_4 generating substrates, specifically the suppression of acetic acid production due to elevated Fe^{2+} levels in biochar-amended soils which further contributes to CH_4 mitigation (Wang et al., 2014).

Biochar application may suppress soil N_2O emissions through several mechanisms. First, it improves soil aeration, which increases oxygen availability and reduces anoxic conditions, thereby limiting the denitrification process. Second, biochar raises soil pH, promoting the conversion of N_2O to N_2. Third, it reduces substrate availability by adsorbing NO_3^- and/or NH_4^+ on its surface, which limits both nitrification and denitrification (Jeffery et al., 2011, 2016; Cayuela et al., 2014). Additionally, biochar can reduce nitrogen leaching, thus helping in reducing nitrate contamination in groundwater (Laird et al., 2010; Bai et al., 2015). The yield improvement of rice with the application of biochar could be associated with the availability and retention of nutrients and water along with better rhizospheric activity which improved the soil fertility (Agegnehu et al., 2016; He et al., 2020). The above-mentioned reasons could lead to lower GWP and GHGI in rice production system under biochar application. Biochar possesses a highly recalcitrant structure and contains stable forms of carbon, which can influence microbial activity by limiting their ability to decompose it efficiently (Ouyang et al., 2014; Nan et al., 2021). This resistance to microbial breakdown slows carbon mineralization and reduces enzymatic processes in the soil (Zimmerman, 2010; Lehmann et al., 2011; Palansooriya et al., 2019). This could make biochar a promising strategy for enhancing carbon sequestration in paddy ecosystems. The high heterogeneity ($I^2 > 90\%$ for most variables) observed in this study indicates that the effects of biochar are strongly context-dependent rather than uniform across all conditions. Such variability is expected in global meta-analyses and likely reflects differences in soil type, climate, cropping systems, biochar feedstock, pyrolysis conditions, and application rates. While the overall direction of effects, including reductions in GHG emissions and improvement in TSCSR and yield, remained consistent, the magnitude of these effects varied widely across studies. Therefore, the results should be interpreted with caution, particularly for site-specific applications.

4.2. Effect of experimental condition on GHG emissions, GHGI, TSCSR, and grain yield

Biochar application reduced N_2O emissions and GHGI, while increasing TSCSR along with rice yield in both field and pot experiments. The reduction in GHGI in these experiments reflects the combined influence of reduced emissions and improved yield, although their relative contributions may vary between experimental conditions. However, it led to an increase in CH_4 emissions and GWP in pot experiments. This pattern may reflect differences between pot and field systems, including experimental scale, duration, and environmental control. Biochar addition under such restricted conditions may further induce microbial shifts, such as stimulation of methanogenic archaea and limited enhancement of methanotrophic activity. Moreover, the lower soil buffering capacity and sharper redox fluctuations in pots can intensify anaerobic conditions compared to paddy field soils, where natural heterogeneity, root dynamics, and larger soil volume moderate these effects, although the specific mechanisms remain uncertain. These mechanisms may vary across experimental setups and should not be interpreted as universally applicable across all conditions. However, some researchers reported that, field or pot experiment not significantly affected yield and GHG emissions under biochar application (He et al., 2017; Liu et al., 2019). These field (<6 years) and pot studies (<5 months) are short term and give ambiguous results therefore it is needed to conduct a long-term trial of biochar to clarify the persistence of its impact on GHG emission, GHGI, TSCSR and crop productivity of rice ecosystem, particularly under field conditions. However, the underlying mechanisms may vary depending on experimental conditions such as system type, environmental factors, and management practices, and should not be considered uniform across all contexts.

Biochar exhibited distinct effects across climatic regions for all key parameters assessed in this study. Tropical regions were associated with relatively greater reductions, with greater reductions in GHG emissions and GHGI. In these regions, the lower GHGI appears to be primarily driven by reductions in GHG emissions, while yield contributions may be secondary. In contrast, subtropical regions recorded higher rice productivity, while temperate regions were associated with greater TSCSR. However, the reported increase in TSCSR should be interpreted with caution, as it reflects changes in total soil carbon following biochar application. This increase likely includes both direct inputs of biochar-derived carbon and changes in native SOC, rather than representing only intrinsic SOC sequestration. It is important to note that these climate-related patterns may partly reflect differences in study

composition, experimental design, and management practices across regions, rather than being driven solely by climatic conditions. The enhanced GHG and GHGI reduction in tropical zones may be influenced to higher temperatures that accelerate organic matter decomposition, where biochar may play a role in stabilizing carbon and improving nutrient retention (Lehmann and Joseph, 2009; Bai et al., 2025). Subtropical climates, with their balanced temperature and precipitation, may favour enhanced microbial activity and nutrient cycling, contributing to yield improvements (Yang et al., 2021). Accordingly, in subtropical regions, reductions in GHGI may be more closely associated with yield enhancement in addition to emission changes. In temperate regions, cooler and wetter conditions may slow microbial decomposition, allowing biochar to persist longer and potentially promoting greater long-term carbon accumulation (Gross et al., 2021; Zimmerman et al., 2011). These region-specific responses suggest the potential value of climate-specific biochar application strategies for maximizing both agronomic and environmental benefits in rice-based systems

In terms of country-specific effects, larger reductions in CH_4 emissions and GWP were observed in studies conducted in China, along with a significant increase in TSCSR, indicating a strong response of biochar application in Chinese rice production systems. In contrast, studies from OTC (countries other than China) reported relatively greater reductions in N_2O emissions and GHGI, along with improvements in rice productivity, highlighting regional variability in the effectiveness of biochar application. The observed reduction in GHGI across regions may result from either emission mitigation or yield improvement, depending on local conditions, and should therefore be interpreted accordingly. However, these regional patterns should be interpreted with caution, as differences in sample size, climatic conditions, soil characteristics, and experimental management among studies may influence the observed responses. It should also be noted that the OTC category includes a wide range of agroecosystems, which may mask important regional differences. Therefore, the regional comparisons presented here primarily reflect descriptive trends and may partly capture underlying differences in study conditions rather than purely geographic effects.

4.3. *Effect of management practices on GHG emissions, GHGI, TSCSR, and grain yield*

Among the cultivation methods, the impact of biochar on reducing CH_4 emissions, GWP, and GHGI, along with enhancing TSCSR and rice yield, was more pronounced under PTR. The lower GHGI under PTR appears to reflect the combined effect of emission reduction and yield enhancement. In contrast, a greater reduction in N_2O emissions was observed under DSR conditions. Under DSR, changes in GHGI may be more strongly influenced by emission reductions, particularly N_2O, rather than yield differences. The higher CH_4 mitigation under PTR may be attributed to biochars role in improving soil aeration, which stimulates CH_4 oxidation and alters microbial dynamics by reducing the methanogenic-to-methanotrophic ratio (Jeffery et al., 2016; Feng et al., 2012). Additionally, the influence of biochar on enhancing mineral-organic interactions and inducing negative priming effects under anaerobic conditions likely promotes long-term carbon stabilization in PTR systems (Liu et al., 2018; Bu et al., 2022). On the other hand, DSR systems are more aerobic and naturally favour nitrification and denitrification cycles, which are the key processes driving N_2O emissions. Therefore, biochars impact on reducing N_2O is more pronounced in DSR than in PTR. This reduction could be associated with biochars capacity to adsorb nitrate (NO_3^-) and ammonium (NH_4^+), thereby limiting substrates for nitrification, and the release of polycyclic aromatic hydrocarbons (PAHs) from biochar that may inhibit nitrifying microorganisms (Cayuela et al., 2014). Hence, under DSR, where alternate oxidation-reduction cycles are more active, the suppression of these pathways by biochar leads to greater N_2O mitigation compared to PTR (Zhang et al., 2012). These mechanisms are strongly influenced by water regime, redox dynamics, and management conditions, and therefore may vary across systems rather than representing a single dominant pathway.

In terms of varied water management practices, AWD was associated with comparatively greater reductions in GHG emissions, GWP, and GHGI, and an increase in TSCSR and grain yield compared with CF. The reduction in GHGI under AWD is likely driven primarily by lower GHG emissions, with yield improvements contributing additionally. However, the opposite is true for CH_4 emission. This pattern may reflect differences in soil redox conditions between CF and AWD systems, where CF environments generally favour CH_4 emissions. The comparatively greater reduction under biochar application in CF systems may be associated with changes in soil microbial activity and rhizosphere processes, although this cannot be directly inferred from the effect sizes alone (Feng et al., 2012; Jeffery et al., 2016). On the other hand, N_2O emission showed greater reduction under AWD, which may be related to biochar-mediated influences on nitrification and denitrification processes under intermittently aerobic conditions (Cayuela et al., 2014).

Along with crop establishment and irrigation regimes, nitrogen application rate also significantly affects the GHG emissions, GHGI, TSCSR, and grain yield. The CH_4 and N_2O emission reduced significantly when nitrogen was applied at a rate of 0–100 kg ha^{-1} (low) but GWP and GHGI were reported to reduced more at medium rate of N application (100–200 kg ha^{-1}). The lower GHGI at medium nitrogen rates reflects a balance between emission reduction and yield response under biochar-amended conditions. The grain yield was higher at high rate of N application (>200 kg ha^{-1}). Both CH_4-oxidizing microorganism (methanotrophs) and the CH_4-producing microorganism (methanogens) require nitrogen for energy and growth (Wu et al., 2019). As previously discussed, biochar mitigates CH_4 emissions primarily by enhancing soil aeration, which promotes methane oxidation and supports the proliferation of methanotrophs (Jeffery et al., 2016; Feng et al., 2012). The stronger mitigation effect observed at 0–100 kg N ha^{-1} may be associated with the lower availability of excess nitrogen, which otherwise stimulates methanogenic activity under anaerobic conditions. Therefore, applying biochar in combination with low nitrogen rates can be an effective strategy to further suppress CH_4 emissions. However, with the increase in the rate of nitrogen fertilizer application, soil N_2O emission generally increases when nitrogen is applied alone (Shcherbak et al., 2014). But when nitrogen is applied along with biochar, the biochar can help to reduce N_2O emissions by influencing key microbial processes such as nitrification and denitrification, which are the primary pathways of soil N_2O production (Cayuela et al., 2014). The surface of biochar adsorbs NO_3^- and/ or NH_4^+, which improves the nitrogen use efficiency (NUE) and may partly explain the observed increases in rice grain yield with higher nitrogen application rates under biochar-amended conditions (Clough et al., 2013). Therefore, with the N application under biochar-amended soil, a significant reduction in GWP and GHGI was observed. The TSCSR increases with rate of application of nitrogen under biochar treatments, which may be because N-fertilization enhances the net primary productivity, which triggers more microbial activity and thus an increase in carbon sequestration in rice-based systems (Du et al., 2020). The observed increase in TSCSR may partly arise from the incorporation of biochar-derived carbon. As most primary studies did not distinguish between biochar-C and native SOC, the reported TSCSR values may overestimate the extent of native SOC sequestration. These process-based interpretations are inferred from existing literature and should be viewed as potential explanations rather than direct evidence derived from this meta-analysis.

The biochar applied at medium rates (10–30 t ha^{-1}) significantly reduced the CH_4 and N_2O emissions, GWP, and GHGI, while enhancing TSCSR and rice grain yield, but a decreasing trend was observed when the rate of biochar application exceeded 30 t ha^{-1}. The reduction in GHGI at medium biochar rates reflects the combined influence of emission mitigation and improved yield performance. The CH_4 emissions were even more effectively reduced when biochar was applied at lower rates (<10 t ha^{-1}), which aligns with previous findings (Yang

et al., 2021; Wen et al., 2025). The observed trend may be attributed to moderate biochar rates (10–30 t ha^{-1}) improving paddy soil physico-chemical properties, such as aeration, water retention, and nutrient availability, while also enhancing microbial functioning (Liu et al., 2019; Gross et al., 2021). In contrast, lower rates (<10 t ha^{-1}) likely promote methanotrophic activity, thereby enhancing CH_4 oxidation without negatively affecting other soil processes (Yang et al., 2021; Wen et al., 2025). However, several studies shown conflicting results particularly for N_2O. For example, Cayuela et al. (2013) and Wen et al. (2025) reported a reduction in N_2O emission, because of biochar application. At higher biochar application rates (>30 t ha^{-1}), the inconsistent or even negative effects observed in some studies may be associated with multiple interacting mechanisms. Excessive biochar inputs can supply surplus labile carbon, which may stimulate methanogenic activity through microbial priming, thereby increasing CH_4 production from paddy fields. Additionally, high biochar rates may lead to nitrogen immobilization, as increased microbial demand for available N, temporarily reducing plant-available nitrogen and potentially lowering grain yield. Additionally, some researchers have stated that, at a high application rate of biochar, its effect on GHG mitigation, improvement in TSCSR, and yield is still unclear may be associated with differences in experimental conditions, management practices, soil properties and biochar properties in each study (Laghari et al., 2015; Liu et al., 2019).

4.4. *Effect of soil properties on GHG emissions, GHGI, TSCSR, and grain yield*

The soil texture is considered one of the important factors which alter gas exchange, water holding capacity and nutrient availability of soil (Silver et al., 2000), and there are studies that show that the effect of biochar on GHG emission and crop growth varies with texture of soil (He et al., 2017). The porosity and pore size distribution of soil can be improved with biochar application, as it is highly porous (Liu et al., 2019). Our results of meta-analysis signify that biochar application in coarse-texture paddy soil results in the reduction of CH_4 emission, GWP, and GHGI more effectively than the fine and medium texture paddy soils over the control. The greater reduction in GHGI in coarse-textured soils appears to be mainly driven by emission mitigation rather than yield differences. When biochar is applied to coarse-textured soils, it mixes well and results in better aeration than medium and fine-textured soil (Liu et al., 2016) and thus enhancing the decomposition of SOC and the activity of methanotrophs (He et al., 2017). Therefore, the application of biochar in coarse-texture soil resulted in a more significant reduction of CH_4 emission than the fine and medium-texture soil. Similar kind of result has been reported by others as well (Liu et al., 2019; Zhang et al., 2012). These mechanisms may interact with soil moisture, redox conditions, and nutrient status, and thus are not universally dominant across soil types. Notably, greater N_2O reduction and yield gains were observed in fine-textured paddy soils, where GHGI changes likely reflect combined effects of lower emissions and higher yield, possibly due to improved aeration and altered rhizosphere conditions following biochar application (Liu et al., 2016). Additionally, the leaching loss of N gets reduced (Bai et al., 2015), nutrient availability improves, thus increasing nutrient use efficiency (Clough et al., 2013) and better crop growth. The impact of biochar on N_2O emission and rice yield is more likely to depend on nutrient availability and microbial activity than the physical properties of paddy soils (Clough et al., 2013).

Considering the TSCSR, it was found that biochar application increases TSCSR more significantly in coarse-textured paddy soils compared to fine- and medium-textured paddy soils. Because the majority of studies did not separate carbon sources using approaches such as isotopic tracing or fractionation, the reported increases should be viewed as an integrated measure of total soil carbon dynamics. These values therefore reflect both external carbon inputs and potential SOC stabilization processes. This pattern may be associated with several factors. Coarse soils typically have lower SOC due to limited clay and silt content, which provide fewer binding sites for carbon stabilization. When biochar is added to rice fields, the relative increase in SOC is more noticeable in these soils (Haefele et al., 2011). In contrast, fine-textured soils stabilize organic carbon more effectively through micro-aggregation, which can reduce the impact of additional carbon inputs. Moreover, biochar mixes more uniformly in coarse-textured soils, enhancing its physical protection and carbon stabilization (Sohi et al., 2010; Liu et al., 2016). As a result, biochar-derived carbon is less likely to displace existing SOC in coarse soils, leading to a greater net increase in soil carbon stock in rice soils (Schmidt et al., 2011). These mechanisms are based on existing literature and should be interpreted cautiously, as the present analysis does not directly quantify carbon stabilization pathways. Highlighting the role of SOC in regulating biochar effects, our study showed that CH_4 emissions, GWP, and GHGI were significantly reduced in paddy soils with medium OC ($\geq$ 0.5% and < 0.75%) and OM ($\geq$ 2% and < 4%). The reduction in GHGI under these SOC conditions is likely associated with lower emissions, while yield contributions may vary across studies. In contrast, N_2O emissions were significantly reduced in soils with medium OC and high OM (>4%). Grain yield improvements were more pronounced in soils with medium OC and low OM (<2%), which is consistent with the findings of Wen et al. (2025), although the magnitude of increase was comparatively lower. Despite these patterns, the mechanisms through which initial SOC and OM levels influence greenhouse gas emissions and rice productivity following biochar application cannot be directly established from the present dataset. The observed increase in CH_4 emissions in high OM soils may be attributed to several potential mechanisms. It is hypothesized that higher availability of labile carbon in such soils could stimulate methanogenic activity. Additionally, increased soil moisture retention associated with high OM, combined with the water-holding capacity of biochar, may promote the formation of anaerobic microsites favourable for CH_4 production. Another possible explanation is that biochar-induced priming effects could enhance organic matter mineralization, thereby increasing substrate availability for methanogenesis. These interpretations remain hypothetical due to limited reporting of key mechanistic variables (e.g., redox potential, dissolved organic carbon, microbial dynamics).

The initial pH of the soil was an important factor that significantly affected the GHG emissions, carbon sequestration, and grain yield, thereby affecting the overall effectiveness of biochar (Liu et al., 2016). The present study indicated that CH_4 emissions, GWP, and GHGI were significantly reduces in biochar-amended alkaline soils, which is consistent with the findings of previous studies (Liu et al., 2019). The decrease in GHGI in these paddy soils is primarily linked to emission reductions, although improved yield may also contribute. Biochar, exerting a liming effect that improves nutrient availability (e.g., P, K), reduces metal toxicity (e.g., Al^{3+}), and improves the soil environment, thereby stimulating microbial activity that stabilizes organic matter and converts labile carbon into more recalcitrant forms. Thus, protecting existing carbon stocks and increasing overall carbon sequestration in acidic soils (Glaser et al., 2002; Jeffery et al., 2011; Zhang et al., 2012). These changes contribute to enhanced crop growth and therefore, the percentage rice yield increased by 14.21%, which is aligned with previous studies (Liu et al., 2019). The liming effect of biochar might influence the CH_4 and N_2O emissions and consumption in acidic soil. The biochar application in acidic paddy soils, increases the pH of soil, which provides a congenial environment for methanotrophic bacteria and N_2O reductase, which contribute to CH_4 oxidation and generation of N_2 respectively (Clough et al., 2013; Jeffery et al., 2016; Pascual et al., 2020). Additionally, Al^{+3} toxicity to methanotrophs is reduced, which thereby lowers the CH_4 emissions (Jeffery et al., 2016). Therefore, the main reason for GWP and GHGI reduction in acidic soil might be biochars liming effect (Liu et al., 2019). These mechanisms are proposed based on prior studies and should be interpreted as potential explanations rather than direct evidence from this meta-analysis. In alkaline

soils, the GWP and GHGI are significantly reduced, which is observed after biochar application in rice fields. The potential mechanisms may include the following: the porous structure of biochar adsorbs CH_4 and NO_3^- and/ or NH_4^+ decreasing substrate availability (Clough et al., 2013; Yaghoubi et al., 2014); modification in soil microbiome offering habitat for beneficial microbes, those involved in carbon cycling and methane oxidation (Xu et al., 2012), and reduction of labile carbon (easily decomposable), thereby microbial respiration decreases which alter CO_2 and N_2O emissions (Zimmerman et al., 2011). However, these proposed mechanisms require validation through further research.

Soil nitrogen status also played an important role in modulating biochar effects. When biochar was applied in soil with very high levels of total nitrogen (>200 kg ha^{-1}), this led to more reduction in GWP and GHGI and an increase in TSCSR were noticed. The reduction in GHGI under high nitrogen conditions is mainly associated with emission mitigation rather than yield gains. However, yield improvement was more in low TN (<100 kg ha^{-1}) soil after the biochar application. In contrast, under low TN conditions, changes in GHGI may be more influenced by yield improvement. In soil with high TN, high N levels can stimulate denitrification, a key source of N_2O emissions. Biochar improves soil aeration and adsorbs NO_3^- and NH_4^+, moderating nitrification and denitrification processes and reducing N_2O emissions, which significantly lowers GWP and GHGI (Cayuela et al., 2014). Moreover, it enhances the aggregate stability and microbial activity, helping to convert labile carbon into more stable forms, thus improving carbon sequestration (Lehmann and Joseph, 2009). These interpretations are inferred from prior experimental findings and are not directly tested within the present study. In N-deficient paddy soil, the greater percentage of yield improvement may be associated with enhanced nutrient retention and gradual N release following biochar application. This may improve the N availability to rice, which leads to greater yield response as compared to already N-rich soils where crop demand may already be near saturation (Clough et al., 2013). These responses may also be linked to improved root development and changes in beneficial microbial activity in paddy soils following biochar application (Xu et al., 2012).

4.5. *Effect of biochar properties on GHG emissions, GHGI, TSCSR, and grain yield*

The pH of biochar played a crucial role in the mitigation of GHG emissions, enhancing TSCSR and yield improvement. In the present meta-analysis, it was found that the carbon sequestration significantly increased with the application of moderately alkaline biochar. The percentage reduction in CH_4 emission, GWP, and GHGI (Supplementary fig.4), along with the percentage increase in rice yield, decreases successively with an increase in biochar pH from acidic (<8) to neutral (8−9) to moderately alkaline (9−10) to highly alkaline (>10) range. The observed variation in GHGI across biochar pH levels reflects changes in both emissions and yield, with emission reduction likely being the dominant factor. Conversely, more N_2O emissions reduction has been observed with highly alkaline biochar. Contrasting to these findings, Liu et al. (2019) showed that GHGI has not been affected by the pH of biochar, and yield significantly increased when biochar pH was > 9. Biochar with a high pH significantly reduces N_2O emissions upon application in paddy fields (Zhang et al., 2020). The pH of biochar varies from 7.72 to 10.4 (9.5 on average), whereas the soil pH varies from 4.1 to 7.9 (6.7 on average). In this situation, biochar can induce a liming effect, especially under acidic soil (Chintala et al., 2014). As discussed earlier, the potential mechanism of CH_4 and N_2O mitigation involves several biochar-mediated effects. However, the relative importance of these mechanisms may vary depending on soil conditions, water regime, and biochar characteristics, and should not be considered universally dominant. Acidic or neutral biochar tends to retain more labile carbon and provides a better habitat for microbes in paddy soils. This enhances the population of methanotrophs and improves soil aeration, both of which reduce CH_4 emissions. Additionally, such biochar can induce negative priming effects, stabilize the native organic carbon and further reduce GHG emissions. It also increases the availability of certain nutrients, such as phosphorus and potassium, enhancing plant growth and yield (Zhang et al., 2020). These mechanisms are suggested based on previous studies and should be interpreted cautiously, as they are not directly evaluated in this meta-analysis. The denitrification is more active at low pH and anaerobic conditions. With the application of highly alkaline biochar, soil pH increases, which inhibits the denitrifying bacteria, leading to a shift in the end product. Additionally, ammonium ions get adsorbed on biochar, which reduces their availability for transformation into N_2O (Zhong et al., 2025).

The feedstock source of biochar has strong regulatory effect on its impact on GHG reduction, TSCSR, and grain yield. In this meta-analysis, it was found that, among the different types of feedstocks, the feedstock with a high C:N ratio significantly reduced the percentage CH_4 and N_2O emissions, GWP and GHGI in rice-based systems. The reduction in GHGI under high C:N ratio feedstock is primarily associated with emission mitigation rather than yield enhancement. The biochar made from low C:N ratio feedstock leads to more CH_4 and N_2O emissions because of increased mineralization intensity of soil nitrogen. By contrast, biochar feedstock with high C:N ratio lowers the mineralization of N and thus mitigate the CH_4 and N_2O emission (Hilber et al., 2012; Zhang et al., 2020). Conversely, feedstock with a low C:N ratio greatly increased the TSCSR, and grain yield as observed in the present meta-analysis. In such cases, any changes in GHGI may be more influenced by yield improvement. Similar trends have been reported in previous studies (Lee et al., 2023). These responses may vary depending on interactions with soil nutrient status and management practices, and therefore should be interpreted as context-specific. This pattern may be associated with greater nutrient availability and enhanced microbial activity, which could support improved crop growth. In contrast, high C:N ratio feedstock results in less microbial nitrogen mineralization (more immobilization), which leads to a greater increase in SOC as reported in a previous study (Lee et al., 2023). Overall, the increase in TSCSR associated with biochar application in rice fields should not be interpreted solely as enhanced native SOC sequestration. Instead, it represents a combined effect of biochar-C addition and changes in TSCSR processes, highlighting the need for more refined assessments in future studies.

The pyrolysis temperature governs the properties of biochar. The biochar produced at low pyrolysis temperature (< 650°C) resulted in a significant reduction in CH_4 emission, GWP, GHGI, and greater yield improvement after application of biochar in rice-based systems. The lower GHGI under low-temperature biochar reflects both emission reduction and yield improvement. But N_2O emissions were more significantly reduced with biochar application, which was produced at high pyrolysis temperature (> 650°C). These findings aligned with the findings of several other studies (Liu et al., 2019; Bai et al.,2025). The mechanisms associated with pyrolysis temperature may differ across environmental and management conditions and should not be generalized as a single dominant pathway. This trend may be attributed to the fact that low-temperature biochar typically retains more labile carbon and volatile organic compounds, which enhance microbial activity, particularly methanotrophs, thereby facilitating CH_4 oxidation (Spokas et al., 2012; Singh et al., 2012). Conversely, high-temperature biochar possesses greater surface area, aromaticity, and redox-active functional groups, which are more effective at adsorbing nitrate and regulating denitrification processes, leading to reduced N_2O emissions (Cayuela et al., 2014). These mechanisms are inferred from prior studies and are not directly tested in this meta-analysis. The biochar produced at low pyrolysis temperature (< 650°C) resulted in an increase in TSCSR, which also aligns with the findings of previous studies (Lee et al., 2023). Generally, as pyrolysis temperature increases, biochar tends to exhibit higher pH, electrical conductivity, ash content, carbon stability, and nutrient levels, while its production yield decreases (Yuan et al., 2015). Thus, biochar produced at lower pyrolysis temperatures may offer a more economical option while still supporting improved crop

performance in rice-based systems (Liu et al., 2019).

4.6. Moderator summary

To provide a coherent interpretation of factors influencing the effect of biochar on TSCSR and crop-related outcomes in rice ecosystems, moderator effects were reorganized into six logical groups.

(1) Cultivation system: Differences between flooded rice systems, aerobic systems, and upland rice systems showed clear variation in effect sizes, with flooded systems often exhibiting stronger responses likely associated with enhanced interaction between biochar and anaerobic soil processes.
(2) Irrigation regime: Water management (CF, AWD, or deficit irrigation) was associated with variation in biochar effects, suggesting the central role of soil moisture in microbial activity and carbon stabilization.
(3) Nitrogen application rate: N-rate strata revealed that biochar effects were most pronounced under moderate to high N inputs, where interactions between mineral nitrogen and biochar surface chemistry may play a role in carbon stabilization and plant-microbe dynamics.
(4) Biochar application rate: Increasing biochar rates generally amplified TSCSR, though subgroup patterns suggested relatively smaller incremental responses at higher application levels; however, this does not indicate a statistically supported saturation effect.
(5) Soil factors: Soil texture, pH, and baseline OC consistently moderated outcomes, with sandy or acidic soils showing stronger beneficial effects, possibly related to differences in carbon retention and buffering capacity.
(6) Biochar properties: Feedstock type, pyrolysis temperature, and biochar stability indices also contributed to between-study variation. Highly aromatic, high-temperature biochar tended to yield stronger sequestration effects but may show variable interactions with soil N cycling.

Study duration was not included as a moderator in the present analysis due to inconsistent reporting and an uneven distribution of observations across duration classes. Nevertheless, temporal effects remain important, as short-term studies may show variable responses, whereas long-term studies may reflect cumulative impacts of biochar on soil physicochemical properties and carbon stabilization. Together, this structured moderator grouping summarizes the overall trends and clarifies how management practices, soil conditions, and biochar characteristics may contribute to the heterogeneity observed across studies.

4.7. Dose-response interpretation of biochar application rate

The meta-regression analysis indicates a positive relationship between biochar application rate and TSCSR, suggesting a gradual increase in response with higher application rates. This pattern supports a continuous dose-dependent trend rather than discrete category-based effects. Although subgroup analysis showed comparatively stronger responses at moderate rates (10–30 t ha^{-1}), the absence of a significant quadratic term indicates no clear statistical evidence of saturation within the evaluated range. Therefore, these subgroup patterns should be interpreted as descriptive trends rather than definitive dose-response relationships.

4.8. Practical and economic constraints of biochar application

In addition to biophysical responses, the large-scale adoption of biochar in rice systems may be constrained by economic and practical considerations, particularly in major rice-producing countries. The production, transport, and field application of biochar often involve substantial costs and logistical challenges, which may limit its feasibility for smallholder farmers. Recent studies have emphasized that, despite the theoretical potential of management interventions to improve productivity and reduce emissions, real-world implementation is frequently constrained by limited infrastructure, resource availability, and competing agronomic priorities, especially in developing regions (Xu et al., 2026). For instance, practices such as residue incorporation are often preferred due to their lower cost and immediate benefits, even though they may increase GHG emissions. In this context, the present findings indicate that moderate biochar application rates (10–30 t ha^{-1}), which frequently showed comparatively stronger responses in rice-based systems, may offer a more feasible balance between agronomic benefits and economic constraints, particularly by limiting input and application costs. Therefore, while biochar shows strong potential for enhancing TSCSR and mitigating emissions, its widespread adoption will likely depend on site-specific feasibility, economic incentives, and supportive policy frameworks within rice-based production systems. These considerations were beyond the scope of the present meta-analysis but are important for translating experimental findings into practical applications.

5. Limitation

A key limitation in the current literature is the lack of consistency in how total soil carbon stock response is quantified following biochar application in rice fields. Many studies measure increases in total soil organic carbon without distinguishing between the carbon inherently present in the added biochar and the additional soil-derived carbon stabilized through biochar-soil interactions. This practice can lead to an overestimation of the true TSCSR, as the measured increase often reflects the presence of externally added recalcitrant carbon rather than newly captured atmospheric carbon. Future research should use standardized approaches, such as isotopic labelling or SOC fractionation, to differentiate biochar-derived carbon from native soil carbon. In addition, the inconsistent reporting on study duration across the selected studies limited our ability to evaluate the temporal effects of biochar, which may differ substantially between short-term and long-term conditions. In addition to these limitations, the GWP estimates in this study are based only on CH_4 and N_2O emissions, while CO_2 fluxes and life-cycle emissions were not included due to their high variability and limited data availability. This may lead to an incomplete assessment of the overall climate impact. Moreover, due to substantial heterogeneity among studies and the limited availability of key mechanistic data, the findings should be interpreted as broad, indicative trends rather than definitive or universally applicable causal relationships.

6. Conclusions

This meta-analysis suggests that biochar application in paddy systems reduces CH_4 emissions by 18.11%, N_2O by 15.54%, GWP by 25.51%, and GHGI by 33.16%, while simultaneously increasing TSCSR by 69.96% and rice yield by 11.13%. The observed increase in TSCSR should, however, be interpreted with caution, as it may partly reflect direct inputs of biochar-derived carbon in addition to changes in native SOC. Additionally, these responses varied with climatic conditions, soil characteristics, management practices, and biochar properties, highlighting the complex interactions governing GHG mitigation and crop productivity. Given the high heterogeneity across studies, these findings should be interpreted as general trends rather than definitive, universally applicable effects. Climatic effects were variable, with subtropical regions showing consistent responses across variables, while tropical regions exhibited stronger reductions in integrated indices (GWP and GHGI). Greater yield responses were generally observed under subtropical climates and in soils with fine texture, acidic pH, medium SOC, and low TN. Biochar derived from high C:N ratio feedstocks, with relatively acidic pH and produced at pyrolysis temperatures below 650

°C, showed comparatively stronger performance. Practices such as PTR, improved nitrogen management, and moderate biochar application (10–30 t ha^{-1}) were associated with enhanced TSCSR and reduced GHG emissions, indicating this range as a practical option for rice-based systems. While the economic feasibility of large-scale adoption was not assessed in this study, the long-term carbon sequestration potential of biochar suggests possible opportunities for its inclusion in climate mitigation strategies, including emerging carbon credit mechanisms reported in recent literature (Gupta et al., 2026). Further long-term and field-scale studies integrating agronomic, environmental, and economic assessments are required to support sustainable implementation across diverse rice-based agroecosystems.

CRediT authorship contribution statement

Alok Sinha: Writing–review & editing, Writing original draft, Methodology, Formal analysis, Data curation, Conceptualization. **C.M. Parihar:** Writing–review & editing, Supervision, Resources, Project administration, Data curation, Conceptualization. **P. Panneerselvam:** Writing–review & editing, Supervision, Project administration, Funding acquisition. **Ayan Sarkar:** Writing–review & editing, Methodology, Formal analysis, Data curation, Conceptualization. **K. Srikanth Reddy:** Writing–review & editing, Methodology, Formal analysis, Data curation, Conceptualization. **Kiranmoy Patra:** Writing–review & editing, Methodology, Formal analysis, Conceptualization. **Sneha Bharadwaj:** Writing–review & editing, Writing–original draft, Methodology. **Madhumita Patra:** Writing–review & editing, Formal analysis. **D.K. Sharma:** Supervision. **S.L. Jat:** Supervision. **S.S. Rathore:** Supervision. **Sunil Kumar:** Supervision, Resources, Project administration. **Sonam R. Sherpa:** Writing–review & editing, Methodology, Formal analysis, Conceptualization. **H.S. Nayak:** Writing–review & editing, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Virender Kumar:** Writing–review & editing, Supervision, Project administration, Conceptualization.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.soilad.2026.100120.

Data availability

Data will be made available on request.

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